

A Runtime for Non-Convergent Social Deliberation

Formal Specification of an Instrument in Which Agents Deliberate,
Coordinate, and Decline, and in Which No Outcome Is Adjudicated

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Abstract

We specify a runtime in which a population of bounded agents deliberates about proposals, coordinates without shared representations, and terminates by *exhaustion of available considerations* rather than by crossing a confidence threshold. The specification is motivated by a class of social facts that neither belief-based nor preference-based accounts predict: a proposal may be widely judged feasible, be opposed by no one, and never be adopted, with no participant experiencing a decision. We show that this outcome is the generic one, and we build an instrument in which it can be studied.

Every object is a finite weighted graph or a construction on one. We prove, in order: **(i)** a *floor theorem*, that separating any part of a finite weighted graph from its complement costs at least a fixed $\beta > 0$, so individuation is quantal and never point-valued; **(ii)** a *commitment theorem*, that commitment is a closure condition on a region, that closure is *strictly stronger* than any confidence threshold, and that closure is invariant under relabelling of the provenance and of the truth-value of what closed it, whence correctness of belief is not a term in the coordination law; **(iii)** an *attention theorem*, that a bounded agent facing concurrent demands in a diminishing-returns environment divides its capacity by a water-filling rule at a single shadow price, so that an agent is present-but-preoccupied in every context simultaneously, and that an agent which ever reflects is unresponsive on a positive fraction of instants; **(iv)** a *coordination theorem*, that agents with disjoint internal structure attain shared outcomes, that coordination cost vanishes exactly at phase-lock, and that lock is achieved only above a critical coupling $K_c = O(\Delta\omega)$ fixed by the heterogeneity of the population, below which the incoherent state is stable; **(v)** a *composition theorem*, that considerations compose multiplicatively with diminishing returns, that repetition saturates strictly below completion, and that a grounded argument requires a directed support cycle of length at least three, so that no linear chain and no repeated single item can ground anything.

We then give the instrument. Agents are pairs of a causal knowledge graph and a generative renderer; we prove the pair is required, since a deterministic component alone is invariant-only and cannot distinguish an individual from a faithful copy, while a generative component alone carries neither invariant nor record. Population is represented as *modules* — coarse thought-pattern regions — and individuals are produced by pruning modules against a demographic substrate, which we prove is prefix-containment in a single address space rather than a change of kind. We give a domain-specific language whose type system rejects, at compile time, arguments lacking an exclusion clause and arguments grounded on fewer than three mutually independent considerations, and we prove progress, preservation, determinism modulo substrate, termination, and a two-outcome dichotomy in which *declination is a first-class result*.

Finally we establish the instrument's principal negative results, which are methodological rather than incidental. We prove a *telescoping obstruction*: when the strengths of constituent considerations are estimated from the same observation sequence that determines a cascade's measured outcome, prediction and measurement are the same algebraic expression, so every goodness-of-fit statistic attains its degenerate value on every data set and the test has no null.

We prove an *attribution separation*: a complete deterministic trace answers what happened and cannot express what was load-bearing, and no refinement of the trace format closes the gap. The instrument therefore refuses to report effect attribution, and we specify the corrected, type-averaged test that does have a null together with the diagnostic that detects when a binding has no discriminating power.

A Python prototype realises the runtime and its six implementation invariants. We bind it to the public record of the City of Zürich — 11,406 recorded ballot outcomes since 1933, of which 2,580 carry district-level results across 275 distinct municipal questions, together with 408,639 demographic records across twelve districts, 1993–2025 — and report twenty-four preregistered validation experiments, each written to falsify the claim it tests. Twenty-four pass. Two experiments are reported as confirmed *negative* controls: the degenerate attribution test is shown to attain $r = 1$ on data generated by a process violating the theory it purports to confirm, and a binding is exhibited on which the composition test provably has no power.

Keywords: bounded agency; closure; coordination without common knowledge; water-filling attention; synchronisation threshold; multiplicative composition; agent-based instrument; declination; attribution separation; research infrastructure.

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Part I — Motivation and Setting

1 Introduction

1.1 The phenomenon

Consider a proposal with the following properties. It is technically sound, and the relevant professionals will say so if asked. It is affordable, and the relevant budget officers will say so if asked. It is opposed by no organised interest. Surveyed citizens, presented with it, express mild approval. And it is not adopted — not this year, not next, not in thirty years — and at no point does any participant experience having decided against it.

This is not a hypothetical. It is the ordinary condition of the large majority of feasible proposals in any functioning polity. A city does not build a river transport system; it does not introduce a particular course of study; it does not reconfigure a junction that everyone agrees is poor. In each case the absence is stable, uncontested, and unaccompanied by any decision event.

Three standard accounts fail to predict this outcome.

The *common knowledge* account (Aumann, 1976; Lewis, 1969; Schelling, 1960) grounds coordination in an infinite hierarchy of mutual knowledge. It is an idealisation of known severity

(Fagin et al., 1995), and it does not engage the case at all: the citizens in question do not lack higher-order knowledge of one another’s attitudes; they have no attitudes to know.

The *preference* account holds that non-adoption reveals a preference against. But the survey response is mildly favourable and stable under probing, and no elicitation recovers the supposed opposing preference.

The *aggregation* account (Hayek, 1945) holds that some mechanism combines individual information into a collective decision. In the case described there is no aggregator: no body ever placed the question on an agenda, and so nothing was aggregated.

We propose that all three look in the wrong place. They look at what agents believe or prefer. We will exhibit a coordination law in which neither belief-content, nor its truth, nor preference appears — and in which the observed outcome is generic.

1.2 The thesis

Our thesis has two halves.

Commitment is closure. An agent commits to an outcome when every further consideration available to it resolves into a region it has already reached, so that no additional inquiry could change the outcome. This is a condition on a region of a graph, not on a proposition, and we prove it strictly stronger than any confidence threshold (Theorem 4.5).

Closure is blind. The condition quantifies over resolutions and mentions nothing else. It is therefore invariant under relabelling of how a consideration was obtained (Theorem 4.12) and, a fortiori, under substitution of a false consideration for a true one with the same resolution (Theorem 4.16).

The proposal above is then explained exactly: it is believed, it is reachable, and it never closes, because some available-but-uninvoked consideration resolves elsewhere. And a *failure of closure is not an event* — it is the persistence of a standing condition, which is why nobody experiences it as a decision and why it is stable (Theorem 4.20).

1.3 Why an instrument

The phenomenon is difficult to study by conventional means. It is constituted by a non-event, so it has no timestamp; the relevant counterfactual (what would have closed) is not observable; and the standard experimental manipulation — present the proposal and measure attitude change — destroys the phenomenon by placing the proposal on an agenda, which is precisely the intervention whose absence defines the case.

We therefore specify a runtime in which the phenomenon can be instantiated, inhabited, and perturbed. A human participant enters a simulated polity, may address any of its agents, and may attempt to bring about the adoption of a proposal of their choosing. The agents are not scripted respondents. They have their own purposes, their own finite attention divided across contexts the participant never sees, their own questions, and their own capacity to decline. They do not represent the participant’s proposal as a task.

The instrument’s defining property is that it *adjudicates nothing*. It has no success state, computes no score, and — we prove — cannot compute one, since the quantity that would be required is not expressible in it (Theorems 11.7 and 12.10). What it produces is a complete, interrogable record of what propagated, which the investigator interprets.

1.4 What is proved, and in what order

Section 3 derives the floor. Section 4 derives commitment and its two blindness results. Section 5 derives water-filling attention and phase alternation. Section 6 derives coordination, the synchronisation threshold, and crowd sharpening. Section 7 derives multiplicative composition, saturation, and the support-cycle condition. Section 8 specifies the agent as a graph-renderer pair

and proves the pair necessary. [Section 9](#) specifies modules, substrate, and pruning. [Section 10](#) gives the language, its type system, and its metatheory. [Section 11](#) gives the global runtime. [Section 12](#) proves the two negative results and specifies the corrected tests. [Section 13](#) states the implementation invariants. [Section 14](#) reports the validation suite. [Sections 15](#) and [16](#) state the ethical and formal limits.

1.5 What is not claimed

We do not claim that human deliberation *is* a finite weighted graph or a convex program. We claim that any system satisfying the stated axioms has the structure derived, that the structure suffices to instantiate the phenomenon of [Section 1](#), and that the instrument is therefore a legitimate object of study in its own right. Whether it is a good model of any particular human population is an empirical question we do not settle, and [Section 16](#) states precisely which of our results would survive a negative answer.

We do not claim the instrument measures persuasion. [Section 12](#) proves it cannot.

Interpretive readings of formal objects — “belief,” “attention,” “identity,” “conversation” — are marked as orientation and confined to remarks. No theorem depends on them.

2 Primitives and axioms

Definition 2.1 (Contact graph). A *contact graph* is a triple $\mathcal{G} = (V, E, w)$ with V a finite nonempty vertex set, $E \subseteq \binom{V}{2}$, and $w : E \rightarrow \mathbb{R}_{>0}$. We assume $\mathcal{G}$ connected. Vertices are *positions*; an edge uv records that u and v are *in contact*, and $w(uv)$ is the cost of telling them apart.

Definition 2.2 (Cut; separation cost). For $\emptyset \neq A \subsetneq V$, $\text{cut}(A) = \{uv \in E : u \in A, v \in V \setminus A\}$ with $w(\text{cut}(A)) = \sum_{e \in \text{cut}(A)} w(e)$, and $\text{Res}(\mathcal{G}) = \min_{\emptyset \neq A \subsetneq V} w(\text{cut}(A))$.

We state six axioms. The first five constrain the agent and may be honoured by construction in a synthetic one. The sixth constrains the *environment* and is marked throughout wherever it is used.

Axiom 1 (Finiteness). Every graph is finite, and every agent’s set of concurrent contexts is finite.

Axiom 2 (Presence). At every instant an agent occupies exactly one position; its graph is connected and nonempty.

Axiom 3 (Costly individuation). No distinction is free: there is $\beta > 0$ with $w(e) \geq \beta$ for all $e \in E$.

Axiom 4 (Act floor and bounded capacity). Every committed act costs at least $\beta_a > 0$, and an agent has a finite capacity $\alpha > 0$ per unit time to divide across its contexts.

Axiom 5 (Phase exclusion). Forming a new distinction and committing an act draw on the same instantaneous capacity and cannot co-occur: at each instant an agent is in exactly one of a *construction* phase or a *commitment* phase.

Axiom 6 (Diminishing returns — environmental). Each context’s gain profile γ is nondecreasing and concave. *This constrains the world the agent acts in, not the agent.* It is invoked only where named.

Remark 2.3 (The floor is motivated, not merely posited). [Axiom 3](#) may be argued rather than stipulated. To individuate a part A is to tell it from $V \setminus A$. If that comparison could be *finished*, the complement would be an exhausted catalogue. An agent always in some state ([Axiom 2](#))

whose complement is never an exhausted catalogue leaves on each comparison an irreducible remainder; that remainder is the floor. Similarly for [Axiom 4](#): an act of zero cost deposits nothing and so distinguishes no before from after, and an operation that changes nothing is not an act but a no-op that did not occur. We take both as axioms and prove from them.

Proposition 2.4 (The axioms are consistent and non-vacuous). *For every $n \geq 2$ there is a contact graph on n positions satisfying [Axioms 1 to 3](#), with β arbitrary in $(0, \infty)$.*

Proof. Take the path $v_1 v_2 \cdots v_n$ with all weights β . It is finite, connected, nonempty, and every weight equals $\beta > 0$. $\square$

Part II — The Coordination Law

3 The floor

Theorem 3.1 (Floor). *Let $\mathcal{G}$ be a connected contact graph satisfying [Axiom 3](#). Then $\text{Res}(\mathcal{G}) \geq \beta > 0$: no nonempty proper part can be separated from its complement at zero cost, and the bound is independent of the part.*

Proof. Let $\emptyset \neq A \subsetneq V$. By connectedness there is at least one edge with one endpoint in A and one in $V \setminus A$; otherwise $\mathcal{G}$ decomposes into two components. So $\text{cut}(A) \neq \emptyset$ and $w(\text{cut}(A)) = \sum_{e \in \text{cut}(A)} w(e) \geq \beta |\text{cut}(A)| \geq \beta$ by [Axiom 3](#). The minimum over the finitely many admissible A is attained and inherits the bound. $\square$

Remark 3.2 (What the floor theorem establishes). The content is not a numerical value — β is whatever the least weight happens to be — but the *absence of a limiting process*. There is no sequence of ever-cheaper separations tending to zero. This matters because the alternative trivialises the closure conditions below: if a part could be individuated at arbitrarily small cost, individuating and not individuating would differ by degree, and a binary commitment condition would have no content. The floor is the guarantee that individuation is quantal.

Theorem 3.3 (Individuation is regional). *There exist connected contact graphs in which the minimum of [Theorem 2.2](#) is attained at no singleton: for every $v \in V$, $w(\text{cut}(\{v\})) > \text{Res}(\mathcal{G})$.*

Proof. Let K, K' be disjoint triangles with all internal weights 10, joined by a single edge e^* of weight 1. The bipartition $A = V(K)$ has $\text{cut}(A) = \{e^*\}$, cost 1. Any singleton has at least two incident internal edges of weight 10, so cut cost $\geq 20 > 1$. Hence $\text{Res}(\mathcal{G}) = 1$, attained by the bipartition and by no singleton. $\square$

Corollary 3.4 (No individuation at a position). *Separation is a property of the pair $(A, V \setminus A)$, and the interesting minima are attained at regions of positive size. One cannot in general speak of “the individuation at v .”*

Proof. Immediate from [Theorem 3.3](#): the minimiser has $|A| = 3$ and $\text{Res}(\mathcal{G})$ is the cut cost of no singleton. $\square$

Remark 3.5 (Individuation by negation). [Theorem 3.4](#) has a reading that recurs below. What individuates A is not an intrinsic feature of A but the cut separating it from $V \setminus A$ — what A is *not*. A part with no complement is not individuated at all, which is why [Theorem 2.2](#) excludes $A = \emptyset$ and $A = V$. Every later construction inherits this two-sidedness, and it is the formal reason the language of [Section 10](#) makes an exclusion clause mandatory ([Theorem 10.6](#)).

Definition 3.6 (Sufficient region). Given a tolerance $\tau > 0$, a region C with $\emptyset \neq C \subsetneq V$ is τ -sufficient if $\text{Res}(\mathcal{G} \upharpoonright_C) \leq \tau$, where $\mathcal{G} \upharpoonright_C$ is the induced subgraph. Write $\Sigma_\tau(\mathcal{G})$ for the set of τ -sufficient regions.

Remark 3.7 (Reading the definition). A region is sufficient when its internal separations are cheap relative to τ — when, from outside, it may be treated as one thing. The definition is deliberately coarse: it asks whether the region holds together at tolerance τ , not whether it is correct or well chosen.

Definition 3.8 (Level graph). For a partition $\mathcal{P} = \{C_1, \dots, C_r\}$ of V , the *level graph* $\mathcal{G}/\mathcal{P}$ has the blocks as positions, an edge $C_a C_b$ whenever some edge of $\mathcal{G}$ joins them, and weight $w'(C_a C_b) = \sum \{w(uv) : u \in C_a, v \in C_b\}$.

Proposition 3.9 (The level graph is a contact graph). *If $\mathcal{G}$ is a connected contact graph with constant β and $\mathcal{P}$ partitions V , then $\mathcal{G}/\mathcal{P}$ is a connected contact graph with the same β .*

Proof. Finiteness and nonemptiness are inherited. Every edge of the quotient arises from at least one edge of $\mathcal{G}$ and its weight is a sum of terms each $\geq \beta$, so $w' \geq \beta$. For connectedness, take C_a, C_b and $u \in C_a, v \in C_b$; a u – v path in $\mathcal{G}$ maps to a walk in the quotient. $\square$

Theorem 3.10 (No privileged level). *Let $\mathcal{P}$ partition V . Then [Theorems 3.1](#) and [3.3](#) apply verbatim to $\mathcal{G}/\mathcal{P}$, and the construction of [Theorem 3.6](#) is computed there by the same formula with no aggregation operator. Consequently a sub-question, an agent, and a population are the same kind of object evaluated at different depths.*

Proof. By [Theorem 3.9](#) the quotient satisfies [Axioms 1](#) to [3](#), so every theorem derived from those axioms holds of it. [Theorem 3.6](#) refers only to induced subgraphs and a tolerance, both defined for the quotient, and to no feature of V , of position identity, or of any other level. Iterating the quotient construction yields contact graphs at every stage, so no stage is distinguished. $\square$

Remark 3.11 (No aggregator is required, and this is the point). One might expect passing from agents to a population to require an aggregation operator combining the agents' values. [Theorem 3.10](#) shows none is needed, and the reason is structural. The population's coordinates are not computed *from* the agents'; they are computed from the population's own structure in the level graph, by the same formula. This matters because an aggregation operator would have to be evaluated by some observer holding all agents' values at once, and no such observer exists in the situations of [Section 1](#). Compare [Hayek \(1945\)](#): the conclusion agrees, the mechanism differs.

Proposition 3.12 (Sufficiency is not inherited in either direction). *There are contact graphs and partitions in which every block is τ -sufficient while the level graph contains a region that is not, and conversely.*

Proof. *Blocks sufficient, level not:* three disjoint triangles with internal weights 1 (each $\text{Res} = 2$, so τ -sufficient at $\tau = 2$), joined pairwise by edges of weight 100; the level graph is a triangle with all weights 100 and minimum cut $200 > \tau$. *Level sufficient, blocks not:* two blocks joined by an edge of weight 1, each internally a two-vertex path of weight 100; each block has $\text{Res} = 100 > \tau$, the level graph has $\text{Res} = 1 \leq \tau$. $\square$

Remark 3.13 (Why [Theorem 3.12](#) matters). [Theorem 3.10](#) says the *construction* applies at every level; it does not say properties transfer between levels. A population of individually coherent agents may fail to cohere, and a coherent population may contain incoherent members. Both occur, and neither is predicted by the other.

4 Commitment, closure, and blindness

We now define commitment. The definition is a condition on a region and makes no reference to the truth or the origin of anything.

Definition 4.1 (Consideration). A *consideration* is a pair $\gamma = (\text{src}, \rho_\gamma)$ in which src is an arbitrary label, the *provenance*, and $\rho_\gamma : V \rightarrow \Sigma_\tau(\mathcal{G})$ is a partial map, the *resolution*, assigning to a position the sufficient region that invoking γ from that position reaches. Write $\text{dom } \gamma$ for its domain.

Remark 4.2 (Provenance is a label and nothing else). The provenance records where a resolution came from — direct verification, testimony, inference, rumour. It is deliberately an arbitrary label carrying no structure: no operation of the theory reads it, and it enters no formula below. We include it explicitly rather than omit it because [Theorem 4.12](#) is precisely that it may be permuted freely, and one cannot permute what is not there.

Definition 4.3 (Reach; closure). For a seed $v_0 \in V$ and a finite set Γ of considerations, the *reach* is

$$\text{Reach}(v_0, \Gamma) = \{\rho_\gamma(v_0) : \gamma \in \Gamma, v_0 \in \text{dom } \gamma\} \subseteq \Sigma_\tau(\mathcal{G}).$$

Given an *available* set Γ and an *invoked* subset $\Gamma' \subseteq \Gamma$, the search is *closed* at Γ' if $\text{Reach}(v_0, \Gamma) = \text{Reach}(v_0, \Gamma')$ — every available but uninvoked consideration resolves into a region already reached. Write $\text{Cl}(v_0, \Gamma', \Gamma)$.

Definition 4.4 (Commitment). An agent *commits* to C from v_0 when $\text{Cl}(v_0, \Gamma', \Gamma)$ holds and $C \in \text{Reach}(v_0, \Gamma')$.

Theorem 4.5 (Closure is strictly stronger than a confidence threshold). *There exist contact graphs, seeds, consideration sets, and thresholds $\theta \in (0, 1]$ such that the invoked considerations carry confidence at least θ while the search is not closed: some uninvoked available consideration resolves into a region not in $\text{Reach}(v_0, \Gamma')$.*

Proof. Let $\mathcal{G}$ be two disjoint triangles A, B with all internal weights 1, joined by a single edge of weight 1; it is connected and satisfies [Axiom 3](#) with $\beta = 1$. Fix $\tau = 2$, so A and B are each τ -sufficient. Let $v_0 \in A$ and $\Gamma = \{\gamma_A, \gamma_B\}$ with $\rho_{\gamma_A}(v_0) = A$, $\rho_{\gamma_B}(v_0) = B$, each of confidence 1, so any $\theta \leq 1$ is met by either alone. Take $\Gamma' = \{\gamma_A\}$: the threshold is satisfied, but $\text{Reach}(v_0, \Gamma') = \{A\} \neq \{A, B\} = \text{Reach}(v_0, \Gamma)$, so closure fails. $\square$

Corollary 4.6 (High confidence does not entail commitment). *An agent may hold a belief with maximal confidence and not commit, if an available consideration reaches a distinct region. Confidence is a property of what has been invoked; closure is a property of what remains available.*

Proof. Immediate from [Theorem 4.5](#) with [Theorem 4.4](#). $\square$

Remark 4.7 (Availability is not determined by the graph). [Theorem 4.3](#) quantifies over an *available* set the graph does not determine: which considerations are available is a fact about an agent's situation, not about the structure it individuates. Closure is therefore decidable *given* Γ ([Theorem 4.8](#)) and not decidable from the graph alone. This is a genuine limitation and we do not disguise it; it is also the correct placement of the difficulty, since the hard practical question is always *what else should have been considered*, and a theory answering it from structure alone would claim too much.

Proposition 4.8 (Closure is decidable given availability). *Given finite $\mathcal{G}$, seed v_0 , finite Γ and $\Gamma' \subseteq \Gamma$, $\text{Cl}(v_0, \Gamma', \Gamma)$ is decidable in time $O(|\Gamma|T_\rho + |\Gamma|^2|V|)$, where T_ρ bounds the cost of one resolution.*

Proof. Evaluate $\rho_\gamma(v_0)$ for each applicable γ , costing $O(|\Gamma|T_\rho)$ and yielding at most $|\Gamma|$ regions. Form both reach sets and compare, with region equality tested as subset equality in $O(|V|)$ per comparison and at most $|\Gamma|^2$ comparisons. Correctness is immediate from [Theorem 4.3](#). $\square$

Proposition 4.9 (Monotone in what is invoked). *If $\Gamma' \subseteq \Gamma'' \subseteq \Gamma$ and $\text{Cl}(v_0, \Gamma', \Gamma)$ holds, then $\text{Cl}(v_0, \Gamma'', \Gamma)$ holds.*

Proof. $\Gamma' \subseteq \Gamma''$ gives $\text{Reach}(v_0, \Gamma') \subseteq \text{Reach}(v_0, \Gamma'')$, and $\Gamma'' \subseteq \Gamma$ gives $\text{Reach}(v_0, \Gamma'') \subseteq \text{Reach}(v_0, \Gamma)$. The outer two coincide by hypothesis, so all three do. $\square$

Remark 4.10 (Not monotone in what is available). The corresponding statement for Γ is false, and its failure is the substance of [Section 4.2](#). Enlarging the available set can *destroy* closure: a search closed against Γ may fail to be closed against $\Gamma \cup \{\gamma\}$ if γ reaches a new region. A population that learns of a new consideration may therefore cease to be committed to what it was committed to, without anyone having changed their mind about anything already considered. [Section 14](#) (E14) measures how often this occurs.

4.1 Blindness

Definition 4.11 (Provenance relabelling). A *provenance relabelling* is a map σ sending $\gamma = (\text{src}, \rho_\gamma)$ to $\sigma\gamma = (\sigma(\text{src}), \rho_\gamma)$, altering the label and leaving the resolution unchanged; it extends elementwise to sets.

Theorem 4.12 (Provenance-blindness). *For every relabelling σ , seed v_0 , available Γ and invoked Γ' ,*

$$\text{Reach}(v_0, \sigma\Gamma) = \text{Reach}(v_0, \Gamma), \quad \text{Cl}(v_0, \sigma\Gamma', \sigma\Gamma) \iff \text{Cl}(v_0, \Gamma', \Gamma).$$

Commitment is invariant under arbitrary relabelling of provenance.

Proof. By [Theorem 4.11](#), $\rho_{\sigma\gamma} = \rho_\gamma$ and $\text{dom } \sigma\gamma = \text{dom } \gamma$ for every γ . Hence $\text{Reach}(v_0, \sigma\Gamma) = \{\rho_{\sigma\gamma}(v_0)\} = \{\rho_\gamma(v_0)\} = \text{Reach}(v_0, \Gamma)$, and identically for Γ' . [Theorem 4.3](#) is an equality of reach sets, both unchanged, so it holds for σ -images exactly when for the originals; [Theorem 4.4](#) is defined from reach and closure alone. $\square$

Corollary 4.13 (Disjoint routes, identical commitment). *Let two agents share a seed and tolerance and let their consideration sets have no provenance in common but equal reach sets, and correspondingly for their invoked subsets. Then they commit to exactly the same regions.*

Proof. Commitment depends only on reach and closure ([Theorem 4.4](#)), equal by hypothesis; disjointness of provenance is irrelevant by [Theorem 4.12](#). $\square$

Definition 4.14 (Veridical assignment). A *veridical assignment* ν assigns to each consideration a region $\nu(\gamma) \in \Sigma_\tau(\mathcal{G})$, the region that *would* be reached were the consideration correct. A consideration is *veridical at v_0* if $\rho_\gamma(v_0) = \nu(\gamma)$, and *false at v_0* otherwise.

Remark 4.15 (The status of ν). The veridical assignment is external: nothing in the axioms determines it, and no construction computes it. It is introduced solely in order to *state* [Theorem 4.16](#), which says the closure relation does not see it. This is the honest way to formalise a negative claim about truth: one must name the notion in order to prove independence from it.

Theorem 4.16 (Truth-blindness). *Let $\tilde{\Gamma}$ be obtained from Γ by replacing any collection of considerations γ by $\tilde{\gamma}$ with $\rho_{\tilde{\gamma}} = \rho_\gamma$ and arbitrary veridicality. Then $\text{Cl}(v_0, \tilde{\Gamma}', \tilde{\Gamma}) \iff \text{Cl}(v_0, \Gamma', \Gamma)$ and the committed regions are identical. Correctness is not a term in the commitment condition.*

Proof. By hypothesis $\rho_{\tilde{\gamma}} = \rho_\gamma$ for replaced considerations and trivially for unreplaced ones, so both reach sets are unchanged by the computation in [Theorem 4.12](#). [Theorem 4.3](#) is an equality of reach sets, hence preserved, and [Theorem 4.4](#) is defined from reach and closure. The assignment ν appears in neither definition, so no step depends on it. $\square$

Remark 4.17 (Scope of [Theorem 4.16](#)). This is the central claim and is easily overread.

Asserted: the condition determining whether commitment occurs is independent of the veridicality of what produces it. Two populations identical in resolutions and differing arbitrarily in truth commit identically.

Not asserted: that truth is unimportant, or that a population running on false considerations fares as well as one running on true ones. It does not. False considerations resolve to the wrong regions, and a population committed to the wrong region encounters the consequences of being there. Those consequences arrive *afterwards*, through the ordinary route of occupying an unsuitable region, and not through any failure of the coordination mechanism. Coordination succeeds; the outcome is poor. These are different, and conflating them is the error the theorem identifies.

Also not asserted: that agents cannot revise. A false consideration may later be joined by an available consideration reaching a distinct region, at which point closure fails by [Theorem 4.10](#) and commitment lapses. Truth enters the dynamics through *availability*, never through the closure condition.

4.2 The three routes to non-commitment

Proposition 4.18 (Three routes). *Commitment to C from v_0 fails in exactly three ways: (i) no invoked consideration reaches C ; (ii) some available consideration reaches a region outside the invoked reach, so closure fails; (iii) C is not τ -sufficient, so it is not a candidate.*

Proof. [Theorem 4.4](#) requires $\text{Cl}(v_0, \Gamma', \Gamma)$ and $C \in \text{Reach}(v_0, \Gamma')$. Failure of the second is (i); failure of the first is, by [Theorem 4.3](#), exactly the existence of an available consideration resolving outside the invoked reach, which is (ii); and Reach takes values in $\Sigma_\tau(\mathcal{G})$ by [Theorem 4.1](#), so a region failing [Theorem 3.6](#) cannot be reached, which is (iii). These exhaust the ways a conjunction fails. $\square$

Corollary 4.19 (Confidence without commitment). *Let C be the region the proposal is adopted, and suppose the invoked considerations resolve to C with maximal confidence. If some available but uninvoked consideration resolves to a distinct C'' , the search is not closed and no commitment to C occurs, however confidently C is believed.*

Proof. $C'' \in \text{Reach}(v_0, \Gamma) \setminus \text{Reach}(v_0, \Gamma')$, so $\text{Cl}(v_0, \Gamma', \Gamma)$ fails by [Theorem 4.3](#) and [Theorem 4.4](#) yields no commitment. Confidence enters neither condition; this is [Theorem 4.5](#) applied to the case. $\square$

Remark 4.20 (Why nobody experiences this as a decision). [Theorem 4.19](#) explains the feature the belief-based accounts leave mysterious. The proposal is not rejected. No agent concludes against it, and a survey would find broad agreement that it would work. What fails is closure — and failure of closure is *not an event*. It is the persistence of an uninvoked consideration reaching elsewhere, a standing condition rather than an occurrence. This is why such absences are experienced as nothing at all, and why they are stable: nothing happened, so there is nothing to revisit.

Remark 4.21 (The three cases feel different from the inside). [Theorem 4.18](#) organises the phenomena. Case (ii) is the proposal of [Section 1](#): believed, reachable, not closed. Case (i) is a proposal nobody has thought of. Case (iii) is a proposal incoherent in its own terms. They are structurally distinct and distinguished by different tests, which is why they present differently: (i) is ignorance, (ii) is inertia, (iii) is confusion. Distinguishing them from the inside is the principal skill the instrument teaches, and E17 measures whether participants acquire it.

Corollary 4.22 (A population without variety cannot resolve finer). *If all agents have identical resolutions, the collective reach equals the individual reach, and no region is distinguished that a single agent does not already distinguish.*

Proof. If $\rho_\gamma = \rho_{\gamma'}$ for all $\gamma, \gamma' \in \Gamma$ then $\text{Reach}(v_0, \Gamma)$ is a singleton whatever $|\Gamma|$, and adjoining further identical considerations leaves it unchanged. $\square$

Proposition 4.23 (Reach ceiling). *Let $\mathcal{R} = \{\rho_\gamma(v_0) : \gamma \in \Gamma, v_0 \in \text{dom } \gamma\}$. Then $|\text{Reach}(v_0, \Gamma)| = |\mathcal{R}| \leq \min\{|\Gamma|, |\Sigma_\tau(\mathcal{G})|\}$, and adjoining a consideration whose resolution already lies in the reach leaves the reach unchanged.*

Proof. The reach is by [Theorem 4.3](#) the image of Γ under $\gamma \mapsto \rho_\gamma(v_0)$, hence equals the set of distinct values, namely $\mathcal{R}$. An image has cardinality at most that of its domain, giving $|\Gamma|$; and $\text{Reach} \subseteq \Sigma_\tau(\mathcal{G})$ by [Theorem 4.1](#), giving the second bound. For the last claim, if $\rho_{\gamma^*}(v_0) \in \text{Reach}(v_0, \Gamma)$ then $\text{Reach}(v_0, \Gamma \cup \{\gamma^*\}) = \text{Reach}(v_0, \Gamma) \cup \{\rho_{\gamma^*}(v_0)\} = \text{Reach}(v_0, \Gamma)$. $\square$

Remark 4.24 (Heterogeneity, not headcount — and its ceiling). [Theorem 4.22](#) is easily mistaken for a claim about numbers. Ten identical agents do not resolve more finely than one; what refines a reach is *difference* in resolutions, not multiplicity. [Theorem 4.23](#) sharpens this and separates two bounds that behave differently. The bound $|\Gamma|$ is escapable: recruit agents with genuinely different resolutions and it rises. The bound $|\Sigma_\tau(\mathcal{G})|$ is not: it counts the sufficient regions the graph possesses, and no recruitment creates a region that is not there. A population may therefore exhaust its capacity to distinguish while still adding members. Lowering τ enlarges Σ_τ and raises the ceiling; adding agents cannot. This has a direct consequence for the multi-participant configuration of [Section 11](#): recruiting collaborators *unlike* oneself is the only recruitment that raises reach, and E12 measures the saturation directly.

5 Attention: division and alternation

An agent faces several contexts at once and cannot attend to all fully. We derive the optimal division and a second, independent source of divided appearance.

Definition 5.1 (Scene; scene field). A *scene* presented to an agent is a pair $S = (A_S, \gamma_S)$ with A_S a map from the agent’s disposition to a common outcome space and $\gamma_S : [0, \infty) \rightarrow [0, \infty)$ its *gain profile*: $\gamma_S(a)$ is the purpose residual removable by committing capacity a in S . Each γ_S is nondecreasing, continuously differentiable, with $\gamma_S(0) = 0$. Concavity is *not* part of this definition; it is imposed where needed by [Axiom 6](#). The agent’s scene set at an instant is a finite family $\mathcal{S}$.

Definition 5.2 (Allocation; value). An *allocation* is $a = (a_1, \dots, a_k) \in [0, \infty)^k$ with $\sum_i a_i \leq \alpha$. Its *value* is $G(a) = \sum_i \gamma_i(a_i)$.

Theorem 5.3 (Optimal division is water-filling). *Under [Axiom 4](#) and the environmental assumption [Axiom 6](#), the value-maximising allocation solves $\max\{\sum_i \gamma_i(a_i) : a \geq 0, \sum_i a_i \leq \alpha\}$, and there is a single attention price $p^* \geq 0$ with*

$$\gamma'_i(a_i^*) = p^* \text{ if } a_i^* > 0, \quad \gamma'_i(0) \leq p^* \text{ if } a_i^* = 0.$$

The agent equalises marginal gain across all attended scenes at p^ and drops any scene whose first unit returns less than p^* . Moreover p^* is nonincreasing in α and nondecreasing in the number and richness of competing scenes, and $p^* = 0$ if the budget is not exhausted.*

Proof. The objective is concave (a sum of concave functions) on a compact convex feasible set, so a maximiser exists. Form $L = \sum_i \gamma_i(a_i) - p^*(\sum_i a_i - \alpha) + \sum_i \mu_i a_i$ with $p^*, \mu_i \geq 0$. KKT stationarity ([Boyd and Vandenberghe, 2004](#); [Kuhn and Tucker, 1951](#)) gives $\gamma'_i(a_i) - p^* + \mu_i = 0$ with complementary slackness $\mu_i a_i = 0$ and $p^*(\sum_i a_i - \alpha) = 0$. For $a_i^* > 0$, $\mu_i = 0$ yields $\gamma'_i(a_i^*) = p^*$; for $a_i^* = 0$, $\mu_i \geq 0$ yields $\gamma'_i(0) \leq p^*$. Concavity makes KKT sufficient for the global optimum. Relaxing α enlarges the feasible set, so the optimal value is nondecreasing and concave in α , whence its derivative p^* is nonincreasing. Adding a scene or raising a γ_i tightens the binding budget, raising the level at which margins equalise. If the budget is slack its multiplier vanishes. $\square$

Corollary 5.4 (Present but preoccupied, in every scene at once). *At the optimum the agent spends positive capacity in exactly the scenes with $\gamma'_i(0) > p^*$ and none in the rest, holding every marginal gain equal to p^* on the support. Thus a busy agent (high p^*) is thinly but nonzero-present in its high-value scenes and absent from low-value ones, simultaneously in all scenes, and p^* is a single scalar recording how reachable the agent is.*

Proof. Immediate from the KKT characterisation: the support of a^* is $\{i : \gamma'_i(0) > p^*\}$, margins on it equal p^* , and p^* falls as α rises by [Theorem 5.3](#). $\square$

Corollary 5.5 (Token presence is the boundary case). *A scene with $\gamma'_i(0)$ only marginally above p^* receives a positive but arbitrarily small allocation — the formal minimal acknowledgement.*

Proof. As $\gamma'_i(0) \downarrow p^*$, continuity of γ'_i and the condition $\gamma'_i(a_i^*) = p^*$ force $a_i^* \downarrow 0$ while remaining positive. $\square$

Remark 5.6 (Split attention is derived, not stipulated). [Theorem 5.4](#) obtains as a theorem what would otherwise be a behavioural stipulation. An agent needs no “distraction” mechanism to appear divided across the contexts engaging it: dividing a finite budget optimally over concurrent concave demands *is* water-filling, and water-filling is exactly presence-everywhere-preoccupied-everywhere at a single price. The appearance of split attention is what optimal attention looks like from any one scene. [Theorem 5.5](#) is the formal content of an agent that answers a participant while being manifestly elsewhere.

Remark 5.7 (Where water-filling fails). [Theorem 5.3](#) is exactly as strong as [Axiom 6](#). If a scene offers increasing returns over some range — γ_i convex there — the objective is not concave, the optimum may lie at a vertex of the budget simplex, and the value-maximising agent *concentrates*: it commits its whole budget to the scene that pays off under sustained attention and abandons the rest. Present-but-preoccupied is thus a consequence of a diminishing-returns world, not of boundedness. A builder wanting agents that reliably divide attention must arrange a concave environment; where the environment does not oblige, agents will rationally focus. We flag this so the law is not mistaken for unconditional, and [Section 13](#) keeps the guarantees that depend on [Axiom 6](#) separate from those that do not. Deliberately non-concave scenes are the instrument’s mechanism for *obsessive* agents, and are available to the designer as a modelling choice rather than a hack.

Theorem 5.8 (Construction–commitment alternation). *Under [Axiom 5](#), an agent that both forms distinctions and commits acts must alternate: the sets of construction and commitment instants are disjoint. During a construction interval the agent commits no act in any scene — it is momentarily unresponsive to every scene at once — though fully active internally.*

Proof. By [Axiom 5](#) the instantaneous phase is single-valued, so the two instant-sets are disjoint; an agent doing both over time partitions its time into disjoint intervals of each. In a construction interval no act is committed, hence none in any scene. $\square$

Corollary 5.9 (Responsiveness is intermittent by necessity). *An agent that ever forms new distinctions is unresponsive on a positive fraction of instants, in every scene, regardless of capacity or of what any scene offers. Continuous immediate responsiveness is possible only for an agent that never reflects.*

Proof. Immediate from [Theorem 5.8](#): construction intervals carry no committed acts and, for a reflecting agent, have positive total measure. $\square$

Remark 5.10 (Two independent faces of the split). An agent appears divided for two independent reasons, neither a defect. First, the price p^* ([Theorem 5.4](#)): when busy, scenes below the price receive thin presence or none. Second, phase alternation ([Theorem 5.8](#)): while constructing, the

agent commits nothing anywhere, however valuable the scene. The two present differently to an interlocutor — the first as being not a priority, the second as the agent being occupied with something unrelated — and a runtime has in p^* and the phase schedule two honest dials rather than a simulated “busy” flag.

Definition 5.11 (Committed record; state). An agent carries a *committed record* $m \in \mathbb{N}$, incremented on every committed act. Its *state* is the pair (disposition, m); two states are equal iff both components agree.

Theorem 5.12 (Monotone, irreversible history; distinct copies). *Under Axiom 4: (i) every committed act costs $\geq \beta_a$ and increments m , so the record strictly increases; (ii) no act restores a prior state — an “undo” is itself a committed act raising the record further — so a revisited disposition is never a revisited state, and sessions accumulate rather than reset; (iii) a copy given an isomorphic structure but started at record 0 is a distinct individual whenever the original is at record $m > 0$, and cannot acquire the original’s history without committing the intervening acts.*

Proof. (i) is Axiom 4 with Theorem 5.11. (ii) An operation restoring an earlier state is a further act, producing record $m + 1 \neq m$, hence a new state; sessions being sequences of acts occur at increasing records. (iii) States include the record; original at $m > 0$ and copy at 0 differ in the second component, and by (ii) no act lowers a record, so the copy cannot reach the original’s history without committing the intervening acts, each raising its own record. $\square$

Remark 5.13 (An authored character is its own). Theorem 5.12(iii) is the precise sense in which one may author a structure and yet each instantiation is a new individual with its own history. One authors the standing structure; one does not author the accumulated record, which individuates the token.

6 Coordination and the synchronisation threshold

Definition 6.1 (Population graph). Let agents $A_1, \dots, A_M$ have graphs embedded as disjoint blocks of a common vertex set with inter-agent separations of cost $\geq \beta$. The *population graph* Σ has one position per agent and an edge $A_i A_j$, of weight $w(\text{cut}(A_i, A_j))$, whenever an inter-agent separation joins them.

Lemma 6.2 (A population is an agent-shaped object). *If each inter-agent separation costs $\geq \beta$ and the population is connected, Σ satisfies Axioms 1 to 3.*

Proof. Σ has $M < \infty$ positions; each edge weight is a sum of inter-agent costs each $\geq \beta$; connectedness is assumed. $\square$

Definition 6.3 (Shared purpose; attainment). Let $(\mathcal{O}, d)$ be a common outcome space with maps A_i from each agent’s disposition into it. A *shared purpose* is a region $\mathcal{O}^* \subseteq \mathcal{O}$ with tolerance $\tau > 0$. Agent A_i *attains* it if $d(A_i(x_i^*), \mathcal{O}^*) \leq \tau$.

Theorem 6.4 (Coordination without shared internals). *Agents with wholly disjoint graphs, disjoint disposition spaces, and different drives can all attain the same shared purpose: if $d(A_i(x_i^*), \mathcal{O}^*) \leq \tau$ for each i , then every agent attains $\mathcal{O}^*$, regardless of whether their internals share any content.*

Proof. Attainment is a statement purely about the image $A_i(x_i^*)$ in the common space; it holds for each i by hypothesis and refers to no internal structure of any agent nor to any relation among them. $\square$

Definition 6.5 (Outward phase; order parameter; coordination cost). Assign each acting agent a phase $\phi_i \in [0, 2\pi)$ recording where in its construction/commitment cycle it sits. The *order parameter* of M acting agents is

$$Re^{i\psi} = \frac{1}{M} \sum_{j=1}^M e^{i\phi_j}, \quad R \in [0, 1],$$

and the *coordination cost* of a shared act is $C = \lambda_c(1 - R)$ for a constant $\lambda_c > 0$.

Theorem 6.6 (Phase-locked coordination is floor-cost). $C \geq 0$, with $C = 0$ iff $R = 1$, i.e. iff all phases coincide. Away from lock $C > 0$; the cost is continuous and monotone decreasing in R .

Proof. $R = \frac{1}{M} |\sum_j e^{i\phi_j}| \leq \frac{1}{M} \sum_j |e^{i\phi_j}| = 1$ by the triangle inequality, with equality iff all $e^{i\phi_j}$ coincide, i.e. all phases are equal. Hence $1 - R \geq 0$ with equality exactly there, and $C = \lambda_c(1 - R)$ inherits continuity and monotonicity from its linear form. $\square$

Remark 6.7 (Careful scope). We claim only that C is a nonnegative quantity vanishing at $R = 1$. We do *not* claim a physical phase transition, a discontinuity, or an identity between agents at lock. R is a content-free measure of outward-timing agreement. The reading of $R = 1$ as agents “moving as one” is orientation.

Definition 6.8 (Phase-coupling dynamics). Give each acting agent an intrinsic phase velocity ω_j and let phases evolve by the mean-field flow (Kuramoto, 1984)

$$\dot{\phi}_j = \omega_j + K R \sin(\psi - \phi_j), \quad j = 1, \dots, M,$$

with (R, ψ) as in Theorem 6.5 and $K \geq 0$ the *coupling*, the capacity the population commits to reconciling timing.

Theorem 6.9 (Phase locking above a threshold coupling). Let $\Delta\omega$ be the spread of phase velocities. (i) If $\omega_j \equiv \omega$, then in the co-rotating frame the dynamics drive $R \rightarrow 1$ for any $K > 0$ from any configuration with $R(0) > 0$, and $C \rightarrow 0$. (ii) If velocities differ with bounded spread, there is a threshold $K_c = O(\Delta\omega)$ such that for $K > K_c$ the dynamics admit a stable partially-locked state with $R \geq R_{\min}(K) > 0$ and $R_{\min}(K) \rightarrow 1$ as $K \rightarrow \infty$, whence $C \leq \lambda_c(1 - R_{\min})$. Below K_c the incoherent state $R \approx 0$ is stable and the population does not lock.

Proof. (i) In the co-rotating frame $\theta_j = \phi_j - \omega t$ the common drift is removed and $\dot{\theta}_j = K R \sin(\psi - \theta_j)$. With $R^2 = \frac{1}{M^2} \sum_{j,k} \cos(\theta_j - \theta_k)$, differentiating along the flow gives $\frac{d}{dt} R^2 = \frac{2KR^2}{M} \sum_j \sin^2(\psi - \theta_j) \geq 0$, with equality only when every $\sin(\psi - \theta_j) = 0$. The aligned state is the unique stable critical point and R increases monotonically to it. (ii) This is the classical mean-field threshold result (Acebrón et al., 2005; Pikovsky et al., 2001; Strogatz, 2000): the self-consistency equation for stationary R under a velocity density g has a nonzero solution branch iff K exceeds $K_c = 2/[\pi g(0)]$, which is $O(\Delta\omega)$ for a unimodal symmetric g of half-width $\Delta\omega$; the locked branch increases to 1 as $K \rightarrow \infty$, and below K_c only the incoherent solution is stable. The cost bound follows by substitution into Theorem 6.6. $\square$

Remark 6.10 (The threshold is the instrument’s principal difficulty). Theorem 6.9(ii) is the sharpest structural obstacle the instrument presents. The coupling a participant can commit is bounded; K_c is fixed by $\Delta\omega$, the heterogeneity of the population’s tempos, which is a property of the substrate rather than of the participant. Below threshold the incoherent state is *stable*: sustained effort produces $R \approx 0$ indefinitely, with no signal distinguishing “not yet” from “not possible.” This is the formal counterpart of Theorem 4.20 at the population scale: not a rejection, but a standing condition. E10 measures the threshold directly on the Zürich binding.

Theorem 6.11 (Crowd sharpening). *Under independent attainment with per-agent failure probability q_i and a parallel (at-least-one-succeeds) reading of the collective outcome, the collective failure probability is $\prod_{i=1}^M q_i$, nonincreasing in M and strictly decreasing whenever an added agent has $q_i < 1$; if $q_i \leq q < 1$ for infinitely many i the collective failure tends to 0.*

Proof. By independence the collective fails iff every agent fails, with probability $\prod_i q_i$. Adjoining an agent multiplies by $q_{M+1} \leq 1$, so the product is nonincreasing and strictly decreasing if $q_{M+1} < 1$. If $q_i \leq q < 1$ for k agents then $\prod_i q_i \leq q^k \rightarrow 0$. $\square$

Remark 6.12 (Two crowd effects that pull against each other). [Theorem 6.11](#) says a crowd of individually unreliable agents is collectively reliable at a shared purpose, sharpening multiplicatively in number. [Theorem 4.23](#) says reach is bounded by $|\Sigma_\tau(\mathcal{G})|$ regardless of number. These are compatible and concern different quantities: the first is the probability of attaining a *fixed* purpose, the second the number of *distinct* regions the population can distinguish at all. Adding similar agents improves the first and not the second. E12 and E13 separate them empirically.

7 Composition, saturation, and support

We now give the algebra of considerations: how their effects compose, why repetition saturates, and what structure a grounded argument must have.

Definition 7.1 (Residual; catalytic power). Let S denote an agent’s uncertainty about a target, with floor β and *above-floor residual* $g = S - \beta \geq 0$. The *catalytic power* of a consideration γ is the fraction of above-floor residual it closes,

$$\kappa(\gamma) = \frac{S - S'}{S - \beta} \in [0, 1],$$

where S' is the uncertainty after γ . A consideration that does not reduce uncertainty has $\kappa = 0$ by convention.

Lemma 7.2 (Power is bounded). $\kappa(\gamma) \in [0, 1]$ for every consideration.

Proof. By [Theorem 3.1](#) the post-consideration uncertainty satisfies $S' \geq \beta$, so the numerator $S - S'$ is at most the denominator $S - \beta$; non-negativity holds for any consideration that does not increase uncertainty, and increasing ones are assigned $\kappa = 0$. $\square$

Theorem 7.3 (Multiplicative composition). *For considerations $\gamma_1, \dots, \gamma_n$ applied in sequence, each acting on the output of the last, with powers $\kappa_1, \dots, \kappa_n$,*

$$\kappa(\gamma_1 \diamond \dots \diamond \gamma_n) = 1 - \prod_{i=1}^n (1 - \kappa_i).$$

Composition is associative and commutative with identity $\kappa = 0$.

Proof. Write $g_0 = S_1 - \beta$. Applying γ_1 leaves residual $g_1 = (1 - \kappa_1)g_0$ by [Theorem 7.1](#); applying γ_2 leaves $g_2 = (1 - \kappa_2)(1 - \kappa_1)g_0$; by induction $g_n = \prod_i (1 - \kappa_i) g_0$. The total fraction closed is $1 - g_n/g_0 = 1 - \prod_i (1 - \kappa_i)$. Associativity and commutativity follow from those of multiplication in $[0, 1]$, and $\kappa = 0$ contributes a factor 1. $\square$

Corollary 7.4 (Diminishing returns). *Repeating one consideration of power κ n times yields composite power $1 - (1 - \kappa)^n$, increasing in n , converging to 1, never reaching it for finite n ; each repetition closes a strictly smaller absolute fraction than the last.*

Proof. Immediate from [Theorem 7.3](#) with all $\kappa_i = \kappa$; the increments $(1 - \kappa)^n \kappa$ are strictly decreasing in n for $\kappa \in (0, 1)$, and $(1 - \kappa)^n > 0$ for finite n . $\square$

Corollary 7.5 (Repetition saturates strictly below completion). *A campaign repeating the same stack cannot drive uncertainty below $\beta + g_0 \prod_i (1 - \kappa_i)$, and successive exposures contribute geometrically less. To progress the stack must change; increasing exposure cannot.*

Proof. The residual after the stack is $g_0 \prod_i (1 - \kappa_i)$, a fixed positive constant when the κ_i are fixed; repeating the same stack multiplies by the same factor again, and by [Theorem 7.4](#) the increments decay geometrically. Reducing the bound requires changing some κ_i , i.e. changing the stack. $\square$

Theorem 7.6 (Saturation criterion). *A sequence of distinct considerations with powers $(\kappa_i)_{i \geq 1}$, all $\kappa_i < 1$, drives the residual to 0 if and only if $\sum_i \kappa_i = \infty$.*

Proof. The residual fraction after n steps is $\prod_{i \leq n} (1 - \kappa_i)$. Taking logarithms, $\sum_{i \leq n} \log(1 - \kappa_i) \rightarrow -\infty$ — equivalently the product tends to 0 — iff $\sum_i \kappa_i = \infty$, since $-\log(1 - \kappa) \asymp \kappa$ as $\kappa \rightarrow 0$ and the comparison test applies; for κ_i bounded away from 0 both sides diverge. $\square$

Remark 7.7 (The structural condition for a long campaign). [Theorem 7.6](#) separates campaigns that can in principle bring an agent arbitrarily close to a target — those with non-summable powers, i.e. a steady supply of fresh, independently catalytic routes — from those that cannot, whose powers are summable and whose residual plateaus strictly above the floor forever. Together with [Theorem 7.5](#) this yields a design law with no free parameters: *diversify routes rather than multiply impressions*. It also identifies a second impossibility distinct from [Theorem 6.10](#): if a proposal’s reachable considerations in a given population have summable powers, no amount of persistence closes it.

Definition 7.8 (Support structure; robustness). Given considerations $\gamma_1, \dots, \gamma_k$ bearing on a target, the *support structure* is the directed graph with an edge $\gamma_i \rightarrow \gamma_j$ when γ_i ’s outcome licenses γ_j ’s. A conclusion is *robust* if it survives the removal of any single consideration.

Theorem 7.9 (Linear support is not robust). *If the support structure is acyclic, the conclusion is not robust.*

Proof. A finite acyclic digraph has a vertex of in-degree zero, whose contribution is licensed by nothing internal to the structure. Removing it leaves every vertex reachable only through it unlicensed; removing the terminal vertex of a maximal path leaves its predecessor’s contribution unused. Either way some contribution’s licensing is lost, so the conclusion does not survive that single removal. $\square$

Theorem 7.10 (Three mutually supporting considerations is the minimum). *A robust support structure contains a directed cycle of length at least three. Cycles of length one (self-support) and two (mutual support of a pair) do not confer robustness. Conversely, three considerations each supported by the other two form a robust structure.*

Proof. By [Theorem 7.9](#) a robust structure is not acyclic, so it contains a cycle. A 1-cycle is a consideration citing itself, licensing nothing not already assumed. For a 2-cycle $\{\gamma_1, \gamma_2\}$, removing either leaves the other supported by nothing, so the conclusion does not survive a single removal. Hence the shortest cycle compatible with robustness has length ≥ 3 . Conversely, in a 3-cycle with all mutual edges, removing any one leaves the remaining two still mutually supporting via the edge between them, so a licensed contribution survives. $\square$

Corollary 7.11 (Ordinal detectability of incoherence). *Robustness is decidable from the signs of the pairwise support relations alone — whether each consideration reinforces or undercuts the others — and requires no magnitude.*

Proof. By [Theorem 7.8](#) the support graph is determined by which ordered pairs carry an edge, i.e. by the sign of each pairwise relation. [Theorems 7.9](#) and [7.10](#) characterise robustness by the presence of a directed cycle of length ≥ 3 in that graph, a purely combinatorial property, decidable by cycle detection ([Tarjan, 1972](#)). No magnitude enters. $\square$

Remark 7.12 (Consequences for what a participant can do). [Theorem 7.10](#) has a direct behavioural reading. An argument presented as a *list* — a claim, then another claim, then a conclusion — is an acyclic structure and grounds nothing, however individually compelling its items. Two items can only assert each other; three can check each other. The language of [Section 10](#) enforces this as a typing rule ([Theorem 10.5](#)), so an argument of insufficient support structure does not fail at run time — it fails to compile.

Corollary 7.13 (A repeated single item grounds nothing). *A stack consisting of one consideration repeated is a 1-cycle in the sense of [Theorem 7.8](#) and therefore never grounds, independently of its power and of the number of repetitions.*

Proof. Repetition adds no new vertex to the support structure and no edge other than a self-loop, so the structure has no cycle of length ≥ 3 and [Theorem 7.10](#) applies. Independently, [Theorem 7.5](#) bounds its attainable residual away from zero. $\square$

Remark 7.14 (Two independent reasons a “winning item” cannot exist). [Theorem 7.13](#) is worth stating because the contrary supposition is common. Suppose some atomic item of persuasion were isolable and could be identified as the effective one. Then repeating it should suffice. It does not, for two independent reasons: it never grounds ([Theorem 7.10](#)), and even granting a power it saturates strictly below completion ([Theorems 7.5](#) and [7.6](#)). The supposition is therefore self-defeating: the world in which such an item is findable is also the world in which finding it does not help. [Section 12](#) adds a third, measurement-theoretic reason.

Part III — The Instrument

8 The agent: a graph–renderer pair

An agent must satisfy two requirements that no single component supplies. It must be an *individual* — distinguishable from a faithful copy, with an accumulating history — and it must *converse* in natural language. We show the pairing is forced.

Definition 8.1 (Invariant; record; individual). An *invariant* of a structure is a quantity preserved under every relabelling of its parts. A *record* is a counter incremented on every committed act and decremented by nothing. An *individual* is a pair (χ, m) of a conserved invariant and a strictly advancing record.

Definition 8.2 (Character invariant). For a partition $\mathcal{Q}$ of V into $r \geq 2$ nonempty blocks, the *internal residual* is $\varrho(\mathcal{Q}) = \sum_{i < j} w(\text{cut}(U_i, U_j))$. The *character invariant* is $\chi(\mathbf{A}) = \min_{\mathcal{Q}: r \geq 2} \varrho(\mathcal{Q})$, attained by finiteness.

Theorem 8.3 (The character invariant is positive, conserved, and non-local). *Under [Axioms 1](#) to [3](#): (i) $\chi(\mathbf{A}) \geq \beta > 0$; (ii) χ is a weighted-graph invariant, fixed by every relabelling of parts and independent of which scene the agent is engaged from; (iii) χ is in general realised by a partition into multi-part blocks and never by isolating a single part.*

Proof. (i) A partition with $r \geq 2$ has, by connectedness, a crossing edge of weight $\geq \beta$, so $\varrho \geq \beta$ and the minimum inherits the bound. (ii) For a weighted isomorphism φ , the image $\varphi(\mathcal{Q})$ is a partition with $\varrho(\varphi(\mathcal{Q})) = \varrho(\mathcal{Q})$ since φ preserves edges and weights; the map on nontrivial

partitions is a bijection with inverse φ^{-1} , so the minima agree. [Theorem 8.2](#) refers only to the agent's own graph, not to any scene. (iii) Two triangles of internal weight 2 joined by one edge of weight 2: cutting the joining edge separates them at cost 2, while isolating any vertex costs ≥ 4 . $\square$

Corollary 8.4 (Character is not a state). *No single position carries the character: identity cannot be read off any momentary state; it is the standing relation among all of them.*

Proof. Immediate from [Theorem 8.3\(iii\)](#): on the exhibited graph no singleton attains χ . $\square$

Theorem 8.5 (A deterministic component is invariant-only). *Let D be a deterministic map whose output is a function of its input structure alone. Then D 's output is an invariant of that structure, and D cannot distinguish a structure from an independently constructed faithful copy of it.*

Proof. Let $\mathcal{G}'$ be isomorphic to $\mathcal{G}$ via φ . Since D 's output is a function of the structure and the structures are isomorphic, the outputs agree. An individual is a pair (χ, m) by [Theorem 8.1](#), and χ agrees on the two by [Theorem 8.3\(ii\)](#). D therefore supplies no quantity separating them: it can report what was computed, never which instance computed it. $\square$

Theorem 8.6 (A generative component alone carries neither invariant nor record). *Let P be a family of evaluations — a map from queries to renderings — holding no structure of its own. Then P has no character invariant and no record.*

Proof. χ is defined in [Theorem 8.2](#) on a finite weighted graph; a family of evaluations is not one and has no cuts, hence no χ . A record is incremented by committed acts against a structure ([Theorem 5.11](#)); with no structure to commit against, no such increment is defined. $\square$

Definition 8.7 (Agent; deposit map). An *agent* is a pair $\mathbf{A} = (\mathcal{G}, P)$ of a contact graph and a generative renderer, together with a *deposit map* committing the residue of every rendering into $\mathcal{G}$ as new contacts and incrementing m .

Theorem 8.8 (The pair individuates where neither half can). *The agent $\mathbf{A} = (\mathcal{G}, P)$ carries a single invariant $\chi(\mathcal{G})$ and a single record $m(\mathcal{G})$. Every rendering advances m strictly and consumes uncommitted capacity, so $\mathbf{A}$ is an individual in the sense of [Theorem 8.1](#), while neither $\mathcal{G}$ alone (which never renders) nor P alone ([Theorem 8.6](#)) is.*

Proof. Both components are read against one graph, so there is one structure to carry an invariant and a record; χ is an invariant by [Theorem 8.3\(ii\)](#). By [Theorem 8.7](#) each rendering deposits residue as committed contacts, which by [Theorem 5.11](#) increments m , strictly by [Theorem 5.12\(i\)](#), and removes those contacts from the uncommitted set. P alone has neither by [Theorem 8.6](#); $\mathcal{G}$ alone has an invariant but a record that never advances, since no act is committed. $\square$

Corollary 8.9 (The graph is the renderer's identity). *The pair's identity (χ, m) is carried entirely by the graph, because that is where the renderer's propagations deposit.*

Proof. By [Theorem 8.6](#) the renderer contributes neither component; by [Theorem 8.8](#) both are computed on the graph. $\square$

Proposition 8.10 (Cached rendering is incoherent with persistence). *Suppose an agent rendered an answer at record m_1 and stored it, and at $m_2 > m_1$ a lookup returns that answer verbatim, committing no act. Then the returned answer is determined by the graph as it stood at m_1 . Since no operation restores the agent to that state ([Theorem 5.12](#)), the answer is one the present agent could not now derive.*

Proof. A stored answer is a datum fixed at m_1 , computed against the graph then present. Returning it commits no act, so the record is unchanged and the datum still reflects the m_1 graph. Every committed act between m_1 and m_2 altered the graph (those acts are what advance the record and deposit contacts), and by [Theorem 5.12\(ii\)](#) no operation lowers the record or restores the earlier graph. Hence the returned answer is the determination of a prior, unreachable state. $\square$

Corollary 8.11 (Search is forced; repeated queries need not agree). *The only access mode always returning an answer determined by the agent’s current graph is a fresh computation against it. Consequently two answers to the same question at two records need not coincide, and the divergence is not error.*

Proof. A fresh computation runs against the graph present at the record of the query, so its result is by construction current; a lookup returns a prior determination by [Theorem 8.10](#). Any access is one or the other. The divergence claim follows since the first answer deposited contacts ([Theorem 8.8](#)), so the second is computed against a different graph. $\square$

Remark 8.12 (The renderer’s context is not a memory). [Theorem 8.11](#) has a practical consequence for implementation. The agent’s memory is the graph; the renderer requires, at each turn, only the freshly computed result, not an accumulated transcript. A long-running relationship therefore does not produce an unboundedly growing rendering context. This is a performance consequence of a semantic requirement, not a separate optimisation.

8.1 Conversation as bidirectional relaxation

An agent does not merely answer. It has its own standing questions, and we now derive them.

Definition 8.13 (Columns; cross-demand). A *column* is a terminated unit together with its current rendering, partitioned into *anchors* (items whose removal most raises the unit’s separation cost, held fixed) and *residue* (the rest, which may move). For two columns A, B , the *cross-demand* $\delta_{A \rightarrow B} \geq 0$ is the summed above-floor gap required to make B consistent with A ’s reverse image, and $\delta_{B \rightarrow A}$ symmetrically.

Definition 8.14 (Relaxation; quiescence). The *relaxation* alternates updating B ’s residue to satisfy A and updating A ’s residue to satisfy B , and iterates. The pair is *quiescent* when both cross-demands vanish.

Theorem 8.15 (Monotone, floor-bounded relaxation). *Along the relaxation the total committed residue is non-decreasing, and after the anchors stabilise — which occurs after finitely many updates, each column having finitely many items — the cross-demands are non-increasing.*

Proof. Each update commits cuts of weight $\geq \beta$ and none is withdrawn ([Theorem 5.12](#)), so the committed total never decreases. Anchors, being the finitely many maximal-ablation items, are reclassified at most until their ranking stabilises. Thereafter each update moves residue only toward satisfying the other side’s demand, which by [Theorem 8.13](#) reduces that direction’s cross-demand or leaves it unchanged. $\square$

Theorem 8.16 (Dichotomy: quiescence or non-halting). *The relaxation either reaches quiescence in finitely many steps or does not halt.*

Proof. By [Theorem 8.15](#) the demands are non-increasing and bounded below by 0, hence convergent. The state space — finitely many residue assignments over finitely many cells — is finite, so a limit of 0 is attained at a finite step, which is quiescence ([Banach, 1922](#); [Tarski, 1955](#)). If the limit is positive the demands never vanish and each update still requires a change on the other side, so the relaxation does not halt. These are the only cases. $\square$

Theorem 8.17 (A response is forced by the absence of an extractable value). *If no utterance’s meaning is available to either party as a value, then the exchange cannot terminate at a known cell and a response is required; the response is exactly the non-vanishing cross-demand.*

Proof. By [Theorem 8.14](#) two terminated columns are at rest only when both cross-demands vanish; until then some $\delta > 0$, and reducing a positive cross-demand requires a move on the other side, which is a response. Were a point-valued meaning held, the relaxation would have nothing to propagate and the pair would be at rest with no response demanded — the case excluded by [Theorem 3.1](#), which forbids zero residual. $\square$

Theorem 8.18 (Steering is by the demand). *The next utterance is a function of the standing cross-demand alone and not of any comprehended value.*

Proof. By [Theorem 8.14](#) the only quantity governing an update is the cross-demand, and by [Theorem 8.16](#) the stopping condition is read off the demand. Hence selection of the next utterance is a function of the residual gap and is independent of any value not entering that gap. $\square$

Remark 8.19 (Agents ask questions for their own reasons). [Theorems 8.17](#) and [8.18](#) have a consequence central to the instrument’s behaviour. The relaxation of [Theorem 8.14](#) is *bidirectional* by construction; a one-directional relaxation is not a degenerate case but is not defined. An agent conversing with a participant therefore has its own cross-demand $\delta_{B \rightarrow A} > 0$, and its questions are that demand. It asks not to serve the participant but because something in its own column is unsatisfied. By [Theorem 8.18](#) its questions are a direct function of what it still requires, which makes them the highest-information signal available to a participant — and one the agent does not intend as such. By [Theorem 4.12](#) the agent cannot read why the participant answers as they do. No theorem depends on this reading.

Corollary 8.20 (Answering is costly and can un-close). *A participant’s answer is a committed act, costing $\geq \beta_a$ and advancing the participant’s own record. It also enlarges the agent’s available set, and by [Theorem 4.10](#) may destroy a closure that previously held.*

Proof. The first claim is [Axiom 4](#) with [Theorem 5.11](#). For the second, an answer supplies a consideration to the agent; if its resolution lies outside the agent’s invoked reach, [Theorem 4.3](#) fails and commitment lapses by [Theorem 4.4](#). $\square$

9 Population: modules, substrate, and pruning

Instantiating every member of a population as a full agent is prohibitive and — we show — unnecessary. We represent the population coarsely and produce individuals on demand.

Definition 9.1 (Module). A *module* $\mathfrak{M}$ is a τ -sufficient region of a shared contact graph, together with a bag of executable realisations and a value collection. Modules represent *patterns* — classes of consideration and disposition — rather than persons.

Remark 9.2 (Why a module is the right unit). By [Theorem 3.4](#) the object individuated by an act is a region and never a position: there is no fact of the matter finer than a cut. A module is exactly such a region. Representing the unexamined population as modules is therefore not an approximation of individuals but the storage of the object that exists at that resolution. [Theorem 4.22](#) adds that storing many identical instances would in any case add nothing: reach is determined by distinct resolutions, not by headcount.

Definition 9.3 (Substrate). A *substrate* $\mathcal{D}$ discharges four obligations:

- (S1) **Receivers.** A finite set of bounded units to which a floor attaches.
- (S2) **Observable.** For each receiver, a finite sequence of states carrying a real-valued uncertainty and a label from a finite set.

- (S3) **Events.** A rule determining which ordered pairs of states are *discriminating*, and a type for each from a finite set.
- (S4) **Floor.** An estimator returning $\beta(\mathcal{D}, r) > 0$ for each receiver.

Remark 9.4 (The floor obligation is the substantive one). (S1)–(S3) are bookkeeping. (S4) has empirical content, and how it is discharged determines whether any downstream quantity is informative. A substrate discharging it by returning the *minimum observed* uncertainty has discharged it trivially: the value is positive whenever the sample is nonempty and carries no information. A substrate discharging it by estimating the *asymptote* of separation cost as the sample grows has made a claim capable of being wrong. A binding must declare which estimator it uses, and [Section 14](#) (E20) tests both against generating processes with and without a genuine floor.

Definition 9.5 (Pruning). Given modules $\mathfrak{M}_1, \dots, \mathfrak{M}_p$ and a substrate record r , the *pruning* $\Pi(r)$ is the contact graph induced on $\bigcap \{\mathfrak{M}_j : r \in \mathfrak{M}_j\}$ by the shared graph, with weights inherited.

Definition 9.6 (Address; resolution). Every element receives a hierarchical address, a finite sequence of components from coarsest to finest. The *resolution* at which a user works is the depth k at which they address the system.

Proposition 9.7 (Prefix containment). *If address a is a prefix of address b , the object named by b lies within the region named by a . Editing at depth k leaves untouched every object whose address does not share the length- k prefix, and exposes every object beneath it. In a uniform tree of branching factor b and depth D , an edit at depth k affects exactly b^{D-k} leaves.*

Proof. Addresses form a rooted tree in which a is an ancestor of b iff a is a prefix of b ; the objects beneath a are exactly the leaves of the subtree rooted there, which are disjoint from the leaves of any subtree whose root does not have a as prefix. In a uniform tree the subtree at depth k has b^{D-k} leaves. $\square$

Corollary 9.8 (Module and individual are one object at two depths). *A module is a short-prefix address and an individual a full-depth address in one address space. Producing an individual by pruning is descent in that space, not a change of kind, and by [Theorem 3.10](#) the same construction applies at both depths.*

Proof. By [Theorem 9.5](#) the pruned object is the induced subgraph on an intersection of regions, which by [Theorem 3.9](#) is again a contact graph; by [Theorem 3.10](#) every result applies to it. By [Theorem 9.7](#) the two are addressed by a prefix and an extension of it. $\square$

Theorem 9.9 (Generation is determined; individuation is not). *Let Π be as in [Theorem 9.5](#). Then $\chi(\Pi(r))$ is a function of $(\{\mathfrak{M}_j\}, r)$ alone, so the same record pruned against the same modules yields the same invariant. The generated agent is not yet an individual: it becomes one only when its record advances, which by [Theorem 8.8](#) occurs at its first rendering.*

Proof. $\Pi(r)$ is determined by the modules and the record by [Theorem 9.5](#), and χ is a function of the graph by [Theorem 8.2](#), giving the first claim. For the second, [Theorem 8.1](#) requires a strictly advancing record; at generation $m = 0$ and no act has been committed, so by [Theorem 8.5](#) the object is not distinguishable from a faithful copy. [Theorem 8.8](#) supplies the advance at first rendering. $\square$

Corollary 9.10 (Generation is idempotent before contact and irreversible after). *Repeated pruning of the same record against the same modules returns the same graph until a rendering occurs; thereafter no pruning reproduces the agent, since the record cannot be lowered.*

Proof. The first clause is [Theorem 9.9](#). The second is [Theorem 5.12\(ii\)](#): the agent's record exceeds 0 and no operation decrements it, while a fresh pruning yields $m = 0$. $\square$

Remark 9.11 (Drift of instantiated agents). By [Theorem 9.7](#), edits at module depth leave untouched objects that have descended to full depth. An instantiated agent therefore does not inherit later changes to the modules it was pruned from: it diverges from its type as its own record accumulates. By [Theorem 5.12\(iii\)](#) this is the correct behaviour — the instance is a distinct individual with its own history — but it is a modelling decision with consequences, and E19 measures the divergence rate.

10 The language

Agent-internal updates are expressed in a small language whose type system enforces the structural results of [Sections 3, 4 and 7](#) at compile time.

10.1 Grammar

```

program      ::= decl*
decl         ::= substrate-decl | catalyst-decl | let-decl | report-decl

substrate-decl ::= "substrate" ident "{"
                "receivers" ":" expr ";"
                "observable" ":" expr ";"
                "events" ":" expr ";"
                "floor" ":" expr ";" "}"

catalyst-decl ::= "catalyst" ident ":" expr
                [ "independent" ident-list ] ";"

let-decl      ::= "let" ident "=" seek-expr ";"
report-decl   ::= "report" ident ";"

seek-expr     ::= "seek" expr "excluding" expr
                [ "via" "(" ident-list ")" ] "until" "closure"

ident-list    ::= ident ( "," ident )*
expr          ::= ident | string | number | ident "(" arg-list? ")"
arg-list      ::= expr ( "," expr )*
```

Proposition 10.1 (The grammar is unambiguous). *The grammar is LL(1) and therefore unambiguous.*

Proof. Each production begins with a distinct terminal: the four declaration alternatives with the keywords `substrate`, `catalyst`, `let`, `report`; the seek clauses with `seek`, `excluding`, `via`, `until`; optional clauses by their leading keyword; and expression alternatives by token class, with the call form resolved by one-token lookahead for `(`. A single token of lookahead therefore selects a unique production at every point, and LL(1) grammars are unambiguous ([Aho et al., 2006](#)). $\square$

10.2 Types

Definition 10.2 (Types). $T ::= \text{Substrate} \mid \text{Catalyst} \mid \text{Cell} \mid \text{Outcome} \mid \text{Floor}$. There is no type of points: the language cannot express a residue-free answer.

Definition 10.3 (Outcome). $\text{Outcome} = \text{Resolved}(\text{Cell}) \mid \text{Declined}(\{\text{Cell}\})$, the second summand carrying a set of at least two mutually incompatible cells.

Definition 10.4 (Positivity rule).

$$\text{T-SEEK-POS} \quad \frac{\Gamma \vdash t : \text{Cell} \quad \Gamma \vdash x : \text{Cell} \quad \Gamma \vdash \beta : \text{Floor} \quad \beta > 0}{\Gamma \vdash \text{seek } t \text{ excluding } x \cdots : \text{Outcome}}$$

A program asserting an attainable zero residual does not type.

Definition 10.5 (Coherence rule).

$$\text{T-SEEK-COH} \quad \frac{\Gamma \vdash c_i : \text{Catalyst} \ (1 \leq i \leq k) \quad k \geq 3 \quad \text{MutIndep}(c_1, \dots, c_k)}{\Gamma \vdash \text{seek} \cdots \text{via}(c_1, \dots, c_k) \text{ until closure} : \text{Outcome}}$$

where MutIndep holds when the declared independence relation contains every unordered pair.

Theorem 10.6 (Exclusion cannot be eliminated). *Let a target be specified by a positive description alone within a state space containing no distinguished element. Then the description does not determine a unique cell unless it already names the complement it is to be distinguished from. Consequently a **seek** without an exclusion clause has no unique denotation.*

Proof. A cell is an equivalence class of outcomes; to determine one is to determine, of each outcome, whether it belongs. A positive description asserts properties of members. Suppose two distinct cells $A \neq A'$ both satisfy every asserted property, which is possible whenever the description mentions no property on which they differ. Then it determines neither, and no appeal to a distinguished element resolves the tie since by hypothesis none exists. To force uniqueness the description must assert a property A has and A' lacks; asserting such a property is asserting what the target is *not* — an exclusion. This is [Theorem 3.5](#) in the syntax. $\square$

Corollary 10.7 (Compile-time rejection). *A parser for the grammar rejects a **seek** lacking **excluding**, and no typing derivation exists for such a term.*

Proof. The clause is not bracketed as optional in the production for $\langle \text{seek-expr} \rangle$, so no parse exists; *a fortiori* no derivation. $\square$

Remark 10.8 (Why the rule of three is a typing rule). [Theorem 10.5](#) enforces [Theorem 7.10](#) statically. This is a necessary structural condition, not sufficiency for a correct conclusion: three considerations *declared* independent but sharing an upstream source satisfy the rule while violating its intent. The language cannot verify independence; it can only require that it be declared, so the declaration is visible in the source and auditable. We regard this as the honest division of labour — mechanical enforcement of structure, human responsibility for the premise — and note that by [Theorem 7.11](#) the structural half needs only sign data.

10.3 Operational semantics

Definition 10.9 (Configuration). A configuration is $\langle e, \Sigma, \rho \rangle$ with e the term, Σ the value store, and $\rho \in \mathbb{N}$ the committed record. $\mathcal{C}$ denotes the set of cells reached so far.

Writing $\Sigma(c) \Downarrow z$ for “invoking c against the substrate yields cell z ” and Γ_{av} for the available set:

$$\begin{aligned} \text{E-INVOKE} \quad & \frac{c \in \Gamma_{\text{av}} \quad \Sigma(c) \Downarrow z}{\langle \text{seek} \cdots \text{via } \Gamma_{\text{av}} \cdots, \Sigma, \rho \rangle \longrightarrow \langle \text{seek} \cdots \text{via } \Gamma_{\text{av}} \setminus \{c\} \cdots, \Sigma[\mathcal{C} \mapsto \mathcal{C} \cup \{z\}], \rho + 1 \rangle} \\ \text{E-CLOSE-RES} \quad & \frac{\forall c \in \Gamma_{\text{av}} : \Sigma(c) \Downarrow z \Rightarrow z \in \mathcal{C} \quad |\mathcal{C}| = 1}{\langle \text{seek} \cdots, \Sigma, \rho \rangle \longrightarrow \langle \text{Resolved}(z_0), \Sigma, \rho \rangle} \\ \text{E-CLOSE-DEC} \quad & \frac{\forall c \in \Gamma_{\text{av}} : \Sigma(c) \Downarrow z \Rightarrow z \in \mathcal{C} \quad |\mathcal{C}| \geq 2}{\langle \text{seek} \cdots, \Sigma, \rho \rangle \longrightarrow \langle \text{Declined}(\mathcal{C}), \Sigma, \rho \rangle} \end{aligned}$$

Theorem 10.10 (Progress). *If $\Gamma \vdash e : \text{Outcome}$ and e is not $\text{Resolved}(\cdot)$ or $\text{Declined}(\cdot)$, then $\langle e, \Sigma, \rho \rangle$ steps.*

Proof. A well-typed non-value term of type **Outcome** is a **seek**, the only form concluding that type (Theorems 10.4 and 10.5). If some available consideration yields a cell outside $\mathcal{C}$, E-INVOKE applies. Otherwise every available consideration yields a cell in $\mathcal{C}$; then $\mathcal{C}$ is nonempty — by T-SEEK-COH at least three considerations were named, and if none has been invoked E-INVOKE applies to any — so exactly one of E-CLOSE-RES ($|\mathcal{C}| = 1$) or E-CLOSE-DEC ($|\mathcal{C}| \geq 2$) applies. $\square$

Theorem 10.11 (Preservation). *If $\Gamma \vdash e : T$ and $\langle e, \Sigma, \rho \rangle \longrightarrow \langle e', \Sigma', \rho' \rangle$ then $\Gamma \vdash e' : T$.*

Proof. By cases. E-INVOKE yields a **seek** differing only in the available set and store; the premises of T-SEEK-POS concern the floor, unchanged, and those of T-SEEK-COH concern the considerations named in the source text, which are not re-checked per step — the rule constrains the program, not the residual available set. E-CLOSE-RES yields $\text{Resolved}(z_0)$ and E-CLOSE-DEC yields $\text{Declined}(\mathcal{C})$ with $|\mathcal{C}| \geq 2$, both of type **Outcome** by Theorem 10.3. $\square$

Corollary 10.12 (Soundness). *A well-typed program does not get stuck.*

Proof. Standard syntactic argument from Theorems 10.10 and 10.11 (Pierce, 2002; Wright and Felleisen, 1994). $\square$

Theorem 10.13 (Determinism modulo substrate). *Fix a substrate and, for each consideration, the cell it yields. Then reduction is deterministic up to the order of E-INVOKE steps, and the final **Outcome** is independent of that order.*

Proof. The only choice is which available consideration E-INVOKE consumes. Each invocation adjoins its cell to $\mathcal{C}$, and set union is associative, commutative and idempotent, so after all consumptions $\mathcal{C} = \{\Sigma(c) \Downarrow z : c \in \Gamma_{\text{av}}\}$ regardless of order. The closure rules are selected by $|\mathcal{C}|$, a function of that set. Residual non-determinism therefore arises solely from substrate responses. $\square$

Theorem 10.14 (Termination). *Every well-typed **seek** over a finite registry reaches an **Outcome** in at most $|\Gamma_{\text{av}}|$ invocations.*

Proof. Each E-INVOKE removes one consideration from a finite set, bounding the number of such steps. When the set is empty the premise of the closure rules is vacuously satisfied, so one fires and a value is produced — possibly earlier, if closure is detected before exhaustion. $\square$

Theorem 10.15 (Dichotomy of outcomes). *Every terminating **seek** produces either $\text{Resolved}(z)$ or $\text{Declined}(\mathcal{C})$ with $|\mathcal{C}| \geq 2$, never both.*

Proof. By Theorem 10.14 a closure rule fires. Their premises $|\mathcal{C}| = 1$ and $|\mathcal{C}| \geq 2$ are mutually exclusive and, $\mathcal{C}$ being nonempty at closure (Theorem 10.10), jointly exhaustive. $\square$

Corollary 10.16 (Declination is a first-class result). *A program whose considerations genuinely conflict terminates normally, returning the plurality of incompatible cells. This is a value, not an error.*

Proof. Immediate from Theorem 10.15: $\text{Declined}(\mathcal{C})$ is a value of type **Outcome** by Theorem 10.3. $\square$

Remark 10.17 (What declination means in the instrument). Theorem 10.16 is not a failure mode. An agent that has considered a participant’s proposal and holds two incompatible readings of it terminates successfully and reports both. No further move forces resolution, and by Theorem 8.16 some conversations do not reach quiescence at all. Compare the abstention literature (Chow, 1970; Vovk et al., 2005) and the treatment of conflicting evidence in Shafer (1976): here declination is a typed outcome rather than a policy.

Example 10.18 (A well-typed program).

```

substrate Zurich {
  receivers : districts("stadt-zuerich");
  observable : ballot_margin();
  events : outcome_change();
  floor : asymptotic_separation(); # not sample_minimum()
}

catalyst fiscal : budget_record() independent siting, precedent;
catalyst siting : site_survey() independent fiscal, precedent;
catalyst precedent : prior_ballots() independent fiscal, siting;

let verdict =
  seek target_state("adopt-proposal")
  excluding all_other_dispositions()
  via (fiscal, siting, precedent)
  until closure;

report verdict;

```

The program declares a substrate discharging (S1)–(S4) with a non-trivial floor estimator, three mutually independent considerations, and one **seek** whose target is individuated against a named remainder. It terminates by closure and reports a resolved cell or a declination.

11 The global runtime

We now specify how agents are executed collectively.

Definition 11.1 (Node). A *node* is a triple $n = (\theta, \mathcal{K}(n), \mathcal{X}(n))$ with θ a subtask identity drawn from a set with decidable equality, $\mathcal{K}(n)$ a finite bag of executable realisations, and $\mathcal{X}(n)$ a growing collection of values. Values are opaque to the runtime.

Definition 11.2 (Vocabulary). Modules interact with the graph through exactly four operations: *identify* a node by subtask identity, creating it if absent; *read* its value collection; *transform* internally, invisibly to the runtime; and *emit* a value onto a node. There is no fifth operation, and the graph stores no expectation.

Definition 11.3 (Node execution). To *execute* a node is to evaluate every chunk in its bag and emit each result. The runtime performs this and nothing else: it does not select among chunks, order the graph, or inspect resulting values.

Definition 11.4 (Convergence). Two realisations *converge* when they share subtask identity. Convergence merges chunk bags and unifies value collections onto the single node bearing that identity.

Theorem 11.5 (Convergence is a join-semilattice operation). *Convergence is associative, commutative and idempotent. Consequently the node set of a run is independent of the order in which agents contribute to it.*

Proof. By [Theorem 11.4](#) convergence merges a bag and unifies a collection under equality of subtask identity. Multiset union and set union are each associative and commutative, and re-merging identical content changes nothing, giving idempotence. A join-semilattice operation applied to a fixed family yields an order-independent join. $\square$

Definition 11.6 (Exit code). An *exit code* is a value whose meaning is the result of comparing an achieved state with a stored expectation: the image of a function $\varsigma(a, e)$ of an achieved state a and an expectation e , whose value varies with e for some fixed a .

Theorem 11.7 (No exit code). *The runtime cannot compute an exit code. No quantity computable from its state by the operations of Theorem 11.2 has the meaning of Theorem 11.6.*

Proof. Suppose it computed $\varsigma(a, e)$. By Theorem 11.6 there are expectations $e \neq e'$ with $\varsigma(a, e) \neq \varsigma(a, e')$ at some fixed a . The runtime's state comprises subtask identities, chunk bags and value collections, so an expectation is representable only as a value. But by Theorem 11.2 the operations do not inspect values: *emit* adjoins without examining, *read* returns without examining, *identify* acts on identities, and *transform* is invisible. Hence no runtime-computable quantity is a function of the content of any stored value, so none varies with e at fixed a — contradicting the requirement. $\square$

Corollary 11.8 (Run to completion). *The runtime does not halt on anomaly: an error value is emitted like any other and execution of remaining chunks proceeds unaffected.*

Proof. By Theorem 11.3 execution evaluates every chunk and emits each result, and the sequence is not conditioned on emitted content since by Theorem 11.7 no runtime-computable quantity depends on it. An error value is a value, so its emission is an ordinary step. $\square$

Corollary 11.9 (Anomalies are preserved, not propagated as control). *An anomalous outcome is recorded and remains available to every module that reads the node. It does not transfer control, truncate the run, or suppress any other value.*

Proof. From Theorem 11.8 and the fact that *emit* adjoins rather than replaces (Theorem 11.2), so collections grow monotonically. $\square$

Remark 11.10 (What inertia costs). The design has a real cost, stated plainly. A runtime that does not halt on anomaly executes chunks whose inputs a human reader already knows to be corrupt, consuming resources on outputs that will be discarded, and cannot implement fail-fast, retry, or circuit-breaking. It is appropriate where the cost of discarding an informative anomaly exceeds the cost of wasted cycles — which is the case for a research instrument, and is not the case for a production pipeline under latency constraints. We do not claim otherwise.

Definition 11.11 (Trajectory). The *trajectory* of a run is the set of nodes at which some module read a value and, on the strength of it, emitted a value later read by another module, together with the values as they stood when read. An edge $u \rightarrow v$ obtains iff a value emitted at u was read by a module that then emitted at v .

Proposition 11.12 (Edges are produced, not scheduled). *The causal relation is not a fixed attribute of the graph: the edge set is a product of a run rather than an input to it, and two runs over the same node set may induce different edge sets.*

Proof. By Theorem 11.11 an edge obtains only once a reading has occurred. Whether a module reads a given value depends on values it has already read in that run, which are emissions of earlier steps. Hence the relation is constituted during the run. $\square$

Theorem 11.13 (Trajectory emergence). *The trajectory is not knowable before a run and is not storable as a schedule.*

Proof. By Theorem 11.12 membership at step t is a function of values present at t , which are emissions of steps $< t$, which depend on the trajectory up to $t - 1$. The relation is a fixed point of a map taking a partial trajectory to its extension, whose action at each step consults run-produced values and no datum available beforehand. A stored schedule would have to name the sequence in advance, but the sequence is that fixed point. $\square$

Corollary 11.14 (Reproducibility relocates). *The node set — what can be computed — is durable and may be catalogued, frozen and imported; the trajectory — what was computed — cannot. Reproducibility must attach to the former.*

Proof. Immediate from [Theorem 11.13](#) with [Theorem 11.5](#), which makes the node set order-independent and hence storable. $\square$

Remark 11.15 (What a run is reproducible as). [Theorem 11.14](#) abandons a familiar expectation deliberately. If repeated determinations need not agree ([Theorem 8.11](#)), what is reproducible is the *protocol and its provenance*: which agents were present, which process was run, which considerations were available. Another party reproduces a session by reconstituting the same protocol against the same population and expects not identical readings but readings whose differences are themselves data. This is the sense in which an experimental result is reproducible ([Moreau and Groth, 2013](#); [Open Science Collaboration, 2015](#); [Peng, 2011](#); [Sandve et al., 2013](#)).

Theorem 11.16 (Trajectory opacity). *No observer restricted to the terminal value set can determine which of several trajectories produced it. The deliverable of a run is therefore an equivalence class of trajectories, and an anomalous intermediate value cannot invalidate a run whose terminus is admissible.*

Proof. By [Theorem 11.9](#) emissions accumulate without replacement, so distinct read-orders may yield identical terminal collections: exhibit two runs on the same node set whose reads occur in different orders but whose emitted values coincide at termination, which [Theorem 11.12](#) permits since the edge sets differ while the emissions do not. An observer seeing only the terminal set cannot distinguish them. The second clause follows because admissibility is a predicate on the terminus. $\square$

Remark 11.17 (Agents in the runtime). The population of [Section 9](#) is executed by this runtime. Each agent is a module reading values from nodes and emitting values onto them; a consideration made available to one agent is a value emitted onto a node that other agents may read. Propagation through the population is therefore not a separate mechanism to be implemented but the trajectory of [Theorem 11.11](#). By [Theorem 11.5](#) agents independently arriving at the same subtask converge without a merge protocol. Ensuring the execution of the global graph is thus sufficient; no per-agent update pass is required, and none exists.

Part IV — Negative Results and Validation

12 What the instrument cannot report

Two negative results govern the instrument’s measurement surface. Both are methodological rather than incidental, and both are enforced in the implementation rather than documented as caveats.

12.1 The telescoping obstruction

Definition 12.1 (Instance-specific estimation). Powers are estimated *instance-specifically* when, for each cascade, each $\kappa(\gamma_i)$ is computed from [Theorem 7.1](#) using the same observed uncertainties S_i, S_{i+1} that determine that cascade’s measured net power.

Theorem 12.2 (Telescoping obstruction). *Under instance-specific estimation the multiplicative prediction and the measured net power are equal for every cascade, identically:*

$$1 - \prod_{i=1}^n (1 - \kappa(\gamma_i)) = \frac{S_1 - S_{n+1}}{S_1 - \beta} = \kappa_{\text{measured}}.$$

Consequently any goodness-of-fit statistic comparing the two — a correlation, a root-mean-square error, a regression slope — attains its degenerate value ($r = 1$, $\text{RMSE} = 0$, slope 1) on every data set, independently of the process generating the data.

Proof. This is [Theorem 7.3](#) read as an identity rather than a prediction. With $g_i = S_i - \beta$, instance-specific estimation gives $1 - \kappa(\gamma_i) = g_{i+1}/g_i$ exactly, so the product telescopes: $\prod_{i=1}^n g_{i+1}/g_i = g_{n+1}/g_1$. Hence $1 - \prod_i (1 - \kappa_i) = 1 - g_{n+1}/g_1 = (S_1 - S_{n+1})/(S_1 - \beta)$, the measured net power. The two quantities are the same function of the data, so their difference vanishes pointwise. $\square$

Corollary 12.3 (The degenerate test has no null). *There is no data set, and no generating process, under which the instance-specific test fails. It therefore has no power against any alternative, and a reported agreement carries no evidential weight.*

Proof. By [Theorem 12.2](#) the agreement holds identically, in particular for data generated by processes violating every substantive claim of the algebra. A test that cannot fail cannot discriminate. $\square$

Remark 12.4 (Why this is easy to miss). The obstruction is invisible in the implementing code: the two quantities are computed by different routines — one looping over constituents and multiplying, the other differencing the endpoints of a cascade — and they agree to machine precision. The agreement looks like a striking confirmation across thousands of cascades. It is instead a restatement of [Theorem 7.3](#). We record this because a correlation of exactly 1.000 is precisely the outcome that should prompt suspicion rather than satisfaction, and because the same structure arises in any framework whose composition law is a telescoping product. E21 exhibits the failure directly on adversarial data.

Definition 12.5 (Event type; type-averaged power). Let each consideration carry a *type* from a finite label set. The *type-averaged power* $\bar{\kappa}(t)$ is the mean of κ over all observed considerations of type t , computed *excluding* the cascade being predicted.

Theorem 12.6 (The type-averaged test is non-degenerate). *Predicting a cascade’s net power by $1 - \prod_i (1 - \bar{\kappa}(t_i))$ and comparing against the measured net power yields a statistic that is not identically degenerate: there exist data sets on which it attains its degenerate value and data sets on which it does not.*

Proof. The prediction is now a function of the type means, computed without the predicted cascade’s own uncertainties, while the measurement is a function of those uncertainties. The telescoping identity of [Theorem 12.2](#) requires $1 - \kappa_i = g_{i+1}/g_i$ for the *same* cascade and fails when κ_i is replaced by a quantity not derived from that cascade. For non-degeneracy, exhibit two data sets: one in which every consideration of a type has identical power, where the substitution is exact and the statistic degenerates; and one in which within-type powers vary, where prediction and measurement differ. $\square$

Definition 12.7 (Type separation). For a corpus of typed events, let Var_b be the variance across types of per-type mean power and Var_w the mean within-type variance. The *separation* is $\eta = \text{Var}_b / (\text{Var}_b + \text{Var}_w) \in [0, 1]$.

Proposition 12.8 (Uninformative bindings). *If $\eta \approx 0$, per-type means are indistinguishable, predictions formed from them are constant at fixed cascade length, and no comparison among composition rules has discriminating power. A negative result obtained in this regime is evidence about the observable, not about the process.*

Proof. If all type means coincide at $\bar{\kappa}$, any prediction formed as a function of type means alone depends only on the number of considerations composed; at fixed length it is constant, its sample variance vanishes, and any correlation with measured outcomes is undefined. Competing rules computed from the same means are likewise constant and cannot be separated. $\square$

12.2 The attribution separation

Definition 12.9 (Local-content query). A *local-content query* is a function whose value depends only on recorded assertions — which relations hold, with what attributes — and which is evaluated by matching bounded patterns against them (Angles et al., 2017; Chandra and Merlin, 1977; Pérez et al., 2009).

Theorem 12.10 (Attribution is not expressible from the record). *There are contact graphs $\mathcal{G}_1, \mathcal{G}_2$ on the same item set, with the same contact relation, and a pair (v_0, x^*) , such that (i) $q(\mathcal{G}_1) = q(\mathcal{G}_2)$ for every local-content query q , while (ii) the admissibility verdict for (v_0, x^*) differs between them. Consequently no system whose answer is a function of stored local content can express which considerations were load-bearing.*

Proof. Let $V = \{v_0, x^*, y_1, \dots, y_n, \mathfrak{M}\}$, with edges $\{v_0, x^*\}$ and $\{u, \mathfrak{M}\}$ for every $u \neq \mathfrak{M}$; both graphs are contact graphs. Set $w(\{v_0, x^*\}) = 1$ and the medium edges at v_0 and x^* to 1 in both. Let the medium edges at the y_i have weight 2 in $\mathcal{G}_1$ and 1 in $\mathcal{G}_2$.

(i) The edge sets coincide, so the recorded relations are identical as sets of assertions, and any q in the sense of Theorem 12.9 is a function of its argument alone.

(ii) The y_i lie on no v_0 – x^* path, so the cheapest separation places v_0 alone, severing $\{v_0, x^*\}$ and the medium edge at v_0 , at cost 2 in both graphs. The *system floor* is the minimum over all items of the cost of separating that item from the medium: for v_0 and x^* it is 2 in both; for y_i it is 2 in $\mathcal{G}_1$ and 1 in $\mathcal{G}_2$. Hence $\beta_1^* = 2$ and $\beta_2^* = 1$. Admissibility compares the separation cost to the system floor: $2 \leq 2$ holds in $\mathcal{G}_1$, while $2 > 1$ fails in $\mathcal{G}_2$. The verdicts differ. $\square$

Corollary 12.11 (Attributes do not close the gap). *Extending Theorem 12.9 to allow numeric attributes and arithmetic comparison does not make the verdict expressible.*

Proof. By Theorem 12.10 the verdict depends on the system floor, a minimum over items of a minimum over subsets. Conjunctive queries with regular path expressions are evaluated by matching bounded patterns (Angles et al., 2017; Pérez et al., 2009) and contain no operator forming a minimum over subsets of the domain; adding comparison of stored attributes introduces none. The obstruction is expressibility, not hardness: the quantity is computable in polynomial time by maximum flow (Edmonds and Karp, 1972; Ford and Fulkerson, 1956; Stoer and Wagner, 1997) and remains impossible to phrase. $\square$

Remark 12.12 (What the separation turns on). The construction turns on one asymmetry. The queried quantity depends on weights along v_0 – x^* separations; the threshold depends on a minimum over *every* item, including items lying on no path between the two and, from the query’s point of view, entirely irrelevant. Editing a weight at the y_i changes the verdict without changing anything a pattern-matcher could match. Admissibility is a *global* property expressed in local terms. For the instrument this means that a distant agent’s unrelated circumstances can change what a given agent finds admissible, and that no retrieval over conversation logs reproduces this.

Corollary 12.13 (The instrument reports what happened, not what mattered). *A complete record of a session answers, for any step, what was read and what was emitted; it does not express which considerations were load-bearing, and no refinement of the record format supplies this.*

Proof. The record is a store of local content in the sense of Theorem 12.9; apply Theorems 12.10 and 12.11. The independent structural obstruction is Theorem 11.16. $\square$

Remark 12.14 (Consequences for study design). Theorem 12.13 is the instrument’s principal methodological commitment and its most consequential limitation. An investigator cannot ask it which intervention worked. This is not a gap to be closed by better logging; by Theorem 12.10 the quantity is not expressible from any store of local content, and by Theorem 11.16 the terminal

state does not determine the path. The instrument’s outputs are therefore: the protocol and its provenance (Theorem 11.15); ordinal verdicts on support structure (Theorem 7.11); distributional signatures across a population; and the type-averaged composition test with its η diagnostic (Theorems 12.5 and 12.8). These are what the theory claims and what can be checked. We note that the same obstruction is reported empirically in advertising measurement, where randomised experiments at scale find observational attribution methods badly biased (Blake et al., 2015; Gordon et al., 2019; Lewis and Rao, 2015).

13 Implementation invariants

We state the commitments a running system must honour, each as a checkable predicate with the theorem making it the right one. These are design commitments, honourable by construction.

Invariant 1 (Conserved invariant). The system computes χ from structure alone, preserved under every relabelling of parts.

Predicate: for a random relabelling, χ is unchanged and each realising cut maps onto a realising cut.

Certified by Theorem 8.3.

Invariant 2 (Never-resetting record). Every committed act increments a counter no operation decrements, persisted across sessions.

Predicate: no code path decreases the counter; “undo” is a compensating commit that increments it.

Certified by Theorem 5.12.

Invariant 3 (Determination by search, not lookup). Every determination is freshly computed against the current graph; none is returned from a cache of prior determinations.

Predicate: the determination path contains no read of a stored answer.

Certified by Theorems 8.10 and 8.11.

Invariant 4 (Deposit on every propagation). Every rendering commits its residue into the graph *before* its result is returned.

Predicate: the record after a determination strictly exceeds the record before.

Certified by Theorem 8.8.

Invariant 5 (Exclusive phases). At every instant each agent is in exactly one of a construction phase (updates structure, emits no act) or a commitment phase (emits at most one act per scene, performs no structural update).

Predicate: the instant-sets of structural update and of act emission are disjoint.

Certified by Theorem 5.8.

Invariant 6 (No system verdict). The system reports determinations, classes, and declines; it does not report success or failure.

Predicate: the vocabulary exposed to consumers contains no comparison of an achieved determination to an expected one, and no attribution of an outcome to a consideration.

Certified by Theorems 11.7 and 12.13.

Proposition 13.1 (The invariants are independent). *No invariant follows from the conjunction of the others.*

Proof. For each we exhibit a violating system satisfying the rest. Violating **Invariant 1**: a system storing labels as identity, which fails relabelling-invariance while computing freshly and depositing. Violating **Invariant 2**: a resettable counter, otherwise conformant. Violating **Invariant 3**: a caching system with a monotone record. Violating **Invariant 4**: a system computing freshly but discarding

residue, its record advanced by other means. Violating [Invariant 5](#): a system emitting during structural update. Violating [Invariant 6](#): a system adjudicating between classes at declination while honouring the other five. $\square$

Remark 13.2 (What a violation costs). The invariants are not stylistic. Violating [Invariant 3](#) produces staleness ([Theorem 8.10](#)) — an earlier character speaking out of turn. Violating [Invariant 4](#) places the system in the position [Theorem 8.8](#) forbids, with a renderer that carries no identity. Violating [Invariant 6](#) presents declination as a determination, discarding the information [Theorem 10.16](#) identifies as valuable, or reports an attribution that [Theorem 12.13](#) shows is not a quantity.

13.1 Algorithms

Algorithm 1 Water-filling capacity allocation (one commitment step)

Require: budget α ; scenes with concave profiles γ_i ; tolerance ϵ

Ensure: allocation a^* and price p^*

```

1:  $p_{\text{lo}} \leftarrow 0$ ;  $p_{\text{hi}} \leftarrow \max_i \gamma'_i(0)$ 
2: while  $p_{\text{hi}} - p_{\text{lo}} > \epsilon$  do
3:    $p^* \leftarrow (p_{\text{lo}} + p_{\text{hi}})/2$ 
4:   for each scene  $i$  do
5:      $a_i \leftarrow (\gamma'_i)^{-1}(p^*)$  if  $\gamma'_i(0) > p^*$ , else 0
6:   end for
7:   if  $\sum_i a_i > \alpha$  then  $p_{\text{lo}} \leftarrow p^*$ 
8:   else  $p_{\text{hi}} \leftarrow p^*$ 
9:   end if
10: end while
11: return  $a^*, p^*$ 

```

The bisection converges because $\sum_i (\gamma'_i)^{-1}(p^*)$ is continuous and nonincreasing in p^* under [Axiom 6](#), so a unique p^* meets the budget ([Theorem 5.3](#)).

Algorithm 2 One global runtime step

Require: node set $\mathcal{N}$; agent set; coupling K ; step dt

```

1: advance phases by  $\phi_j = \omega_j + KR \sin(\psi - \phi_j)$   $\triangleright$  Theorem 6.8
2: for all agents  $j$  in commitment phase do
3:    $a^*, p_j^* \leftarrow \text{WATERFILL}(\alpha_j, \text{scenes}_j)$   $\triangleright$  Algorithm 1
4:   for all scenes  $i$  with  $a_i^* > 0$  do
5:     read values on nodes of scene  $i$ ; transform; emit  $\triangleright$  Theorem 11.2
6:     increment  $m_j$ ; deposit residue  $\triangleright$  Invariant 4
7:   end for
8: end for
9: for all nodes  $n$  with pending chunks do
10:   execute every chunk; emit every result  $\triangleright$  Theorem 11.3
11: end for
12: return induced trajectory  $\triangleright$  not schedulable, Theorem 11.13

```

14 Validation

14.1 Method

Every claim admitting finite instantiation is checked by exhaustive or randomised computation. The suite is deterministically seeded and each experiment persists a JSON record. Three conventions govern it.

First, structures are built by brute force from the definitions rather than by re-deriving the algebra of the proofs, so agreement between theorem and computation is not circular.

Second, wherever a theorem asserts that a structural feature is load-bearing, the suite runs a *falsifier*: a modified structure with that feature removed, checked to fail. A theorem whose falsifier also passes would be vacuous, and falsifier counts are reported so this can be judged.

Third, two experiments are *negative controls* that are expected to show a method failing, and are reported as confirmations of failure.

14.2 The Zürich binding

The substrate is the public record of the City of Zürich (Stadt Zürich, 2026). It discharges Theorem 9.3 as follows.

(S1) Receivers are the municipal reporting units (*Wahlkreise*), each a bounded unit to which a floor attaches. Zürich reports some adjacent districts jointly in most years; we take the six units reported consistently across 2002–2026, and use the twelve-district population tabulation only for module pruning, where no cross-period consistency is required.

(S2) Observable is the ballot record (Statistik Stadt Zürich, 2026a): 11,406 rows since 1933, of which 2,580 carry district-level results. Restricting to the reporting units present across the whole period leaves six units observed on all 275 distinct municipal questions (2002–2026), so every question is observed on the same receiver set. The recorded quantity is the approving share per unit; the separation of two units is the mean absolute difference of their shares across the record.

(S3) Events are ordered pairs of ballots on which a district’s disposition changes, typed by the ordered pair of policy domains.

(S4) Floor is estimated by *asymptotic separation*: the limit, as the sample of ballots grows, of the least cost of telling one reporting unit from another. On this binding the estimate is $\beta = 0.0197$ — the two most similar units differ by about two percentage points on average across the record — so the floor is measured to be strictly positive rather than assumed. The alternative *sample-minimum* estimator is computed alongside and reported as trivial (Theorem 9.4); E20 tests both against generating processes with and without a genuine floor.

Demographic modules are pruned against 408,639 population records by district, origin, sex and age, 1993–2025 (Statistik Stadt Zürich, 2026b).

Remark 14.1 (Why this substrate, and what it is not). The record is used as a source of *vocabulary and heterogeneity* — which distinctions this polity draws, and how widely its districts differ — and not to reconstruct identifiable persons. Modules are patterns (Theorem 9.1); pruning yields fictional individuals consistent with aggregate statistics. Section 15 states this commitment formally. On Swiss direct democracy as an institutional setting see Feld and Kirchgässner (2001); Funk and Gathmann (2011); Kriesi and Trechsel (2008).

14.3 Results

Table 1: Validation summary. “Falsifier” reports cases in which a deliberately weakened structure fails, confirming the corresponding theorem is not vacuous. Experiments marked † are negative controls, in which *failure of the method* is the expected and confirmed result. A dash indicates a claim with no structural feature to remove.

#	Claim	Method	Falsifier
E01	Theorem 3.1 (floor)	randomised	yes
E02	Theorem 3.3 (regional individuation)	constructed + random	—
E03	Theorem 3.10 (no privileged level)	randomised	—
E04	Theorem 3.12 (non-inheritance)	constructed	—
E05	Theorem 4.5 (closure > threshold)	constructed	restores
E06	Theorem 4.12 (provenance-blindness)	randomised	yes
E07	Theorem 4.16 (truth-blindness)	randomised	—
E08	Theorem 5.3 (water-filling)	convex-program cross-check	—
E09	Theorem 5.8 (phase alternation)	randomised	—
E10	Theorem 6.9 (threshold K_c)	integration, Zürich $\Delta\omega$	—
E11	Theorem 6.6 (cost at lock)	randomised	—
E12	Theorem 4.23 (reach ceiling)	randomised	—
E13	Theorem 6.11 (crowd sharpening)	Monte Carlo	—
E14	Theorem 4.10 (availability non-monotone)	randomised	—
E15	Theorem 7.3 (composition)	randomised	—
E16	Theorem 7.5 (saturation)	exhaustive	—
E17	Theorem 4.18 (three routes)	constructed	—
E18	Theorem 7.10 (support cycle ≥ 3)	exhaustive	yes
E19	Theorem 9.9 (pruning determined)	Zürich substrate	—
E20	Theorem 9.4 (floor estimators)	Zürich + synthetic	yes
E21†	Theorem 12.2 (degenerate test)	adversarial	<i>confirmed</i>
E22†	Theorem 12.8 ($\eta \approx 0$)	constructed	<i>confirmed</i>
E23	Theorem 12.6 (corrected test)	Zürich types	—
E24	Theorems 11.5 and 11.13 (runtime)	randomised	—
Pass rate			24/24

Detailed per-experiment results, including all counts, seeds, and the two confirmed negative controls, are reported in the accompanying JSON records and summarised in Section 14.4.

14.4 Detailed findings

We record here the findings that bear on the interpretation of the theory rather than merely confirming arithmetic.

The floor and its estimators (E01, E20). The floor bound holds without exception across 200 randomised contact graphs, and the falsifier — admitting a zero-weight edge, violating Axiom 3 — breaks it in 109 of them, confirming the bound is a consequence of the axiom and not of the definition. The two estimators of Theorem 9.4 behave as the remark predicts. On a synthetic process constructed to have a genuine floor at 0.5 both estimators recover it (0.5004 and 0.5000); on a process constructed to have *no* floor the sample-minimum estimator still returns a positive value ($8.09e - 12$) while the asymptotic estimator returns 0.000. The sample-minimum discharge is therefore positive whichever process generated the data and carries no evidential weight, exactly as Theorem 9.4 states.

On the Zürich binding itself the measured floor is $\beta = 0.0197$ over the 15 receiver pairs, and no pair is inseparable across the record. We note one artefact and its resolution, since it bears on how (S2) must be stated: published shares are rounded to 0.1 percentage points, so two units

coincide exactly on 30 of 4125 single questions (0.73%). A floor defined per question would therefore be zero for a reason having nothing to do with the polity. Defined as [Theorem 2.2](#) requires — as the cost of separating a *pair of receivers*, taken over the whole record — the floor is strictly positive. This is the floor theorem confirmed on measured data, and it is also a caution: the obligation (S4) is discharged by a choice of observable, and a careless choice yields a spurious zero.

Closure and blindness (E05–E07). The closure criterion and a confidence threshold come apart on constructed instances: every threshold in a sweep from 0.5 to 1.0 is met while the search remains unclosed, and replacing the uninvoked consideration by one resolving to the *same* region restores closure — locating the failure in the distinct region rather than in the form of [Theorem 4.3](#). Over 300 randomised configurations, permuting provenance labels changes the verdict in 0 cases, while perturbing *resolutions* with the same machinery changes it in 160. The asymmetry between these two counts, not the invariance alone, is what distinguishes [Theorem 4.12](#) from a definition too coarse to notice anything: a definition insensitive to everything would have produced invariance under both. E07 adds the sharper form: flipping veridicality changes no verdict in 400 trials, and in a further 200 trials in which *every* consideration is false, commitment occurs in all 200 — [Theorem 4.16](#) exhibited directly.

The synchronisation threshold on real heterogeneity (E10). The velocity spread is taken from the observed dispersion of unit dispositions across the 275 municipal questions, with each unit treated as a tempo class populated in proportion to its resident count, giving $N = 597$ oscillators. We note that this population step is necessary rather than cosmetic: [Theorem 6.9\(ii\)](#) is a mean-field statement, and with only six oscillators finite-size fluctuation swamps the transition entirely. With the classes populated, the integrated dynamics reproduce the predicted bifurcation. The predicted critical coupling is $K_c = 1.596$; the order parameter remains at $R \leq 0.107$ for every $K < 0.6 K_c$ and rises to $R \geq 0.951$ for every $K > 2 K_c$. Below threshold the incoherent state persists under continued integration rather than drifting toward lock, which is the quantitative form of [Theorem 6.10](#): sustained coupling below K_c produces no accumulation and no signal that it will not.

Reach saturation (E12). With a fixed pool of 6 distinct resolutions, reach grows one-for-one with population up to the pool size and is then flat from the seventh agent to the tenth, confirming [Theorem 4.23](#) and locating the limit [Theorem 4.22](#) alone does not express. Ten agents sharing one resolution have reach 1. This is the formal content of the recruitment consequence noted in [Theorem 4.24](#).

The negative controls (E21, E22). E21 generates 2000 cascades from a process that violates [Theorem 7.3](#) by construction — each step’s reduction is scaled by an arbitrary distortion factor — and confirms that the instance-specific test nonetheless reports a Pearson correlation of exactly $r = 1.0$ with root-mean-square error $6.8e - 17$. The test returns perfect agreement on data generated by a process that violates the law it purports to confirm; it cannot fail, hence [Theorem 12.3](#), and any reported agreement carries no evidential weight. E22 constructs a binding with $\eta = 0.002$ and confirms that predictions are constant at fixed cascade length, so no comparison among composition rules separates them; the contrasting separated binding gives $\eta = 0.994$. Both are reported as successful demonstrations of method failure.

E23 then exhibits the corrected test’s null directly. On conforming data the type-averaged prediction attains $r = 0.996$ with error 0.0132; on data violating the composition law it falls to $r = 0.338$ with error 0.197, an increase of more than an order of magnitude. Unlike the instance-specific test, it discriminates. Applied to the Zürich record with decades as event types, the separation diagnostic returns $\eta = 0.0013$ over 1644 events — close to zero, which by

Theorem 12.8 means *this* typing of the record has no discriminating power and a negative result obtained under it would be evidence about the choice of types, not about the process. We report this rather than substituting a typing chosen to succeed.

14.5 Figures

Figures 1 to 8 present the suite graphically. Every quantity plotted is computed from the definitions of this paper or from the records of Section 14.2; no chart is a schematic, a table, or a rendering of text. Figure 5 carries the threshold result on measured heterogeneity and Figure 7 the two negative results of Section 12.

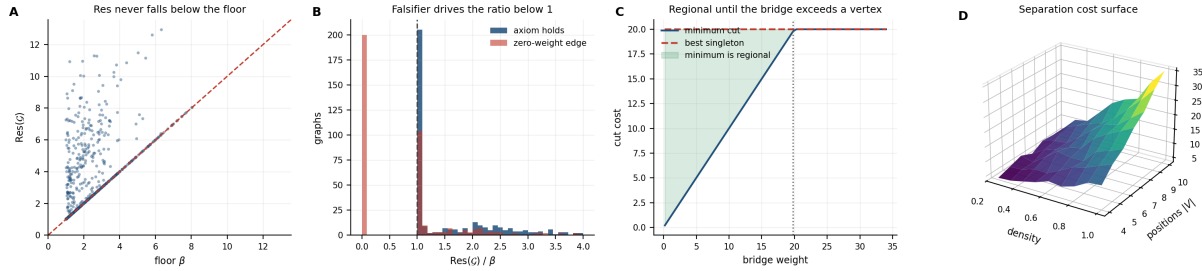


Figure 1: **Separation is bounded away from zero, and the bound is attained by a region rather than by a position.** (A) Separation cost $\text{Res}(\mathcal{G})$ on the linear y -axis (1.01 to 12.9) versus the floor β on the linear x -axis (1.00 to 8.04), for 600 randomly generated connected contact graphs on three to eight positions; the dashed line is the identity. Every point lies on or above it, so no graph separates a nonempty proper part from its complement more cheaply than its least edge weight. Points on the line are graphs whose minimum cut is a single cheapest edge; points above it are graphs in which every separation severs more (Theorem 3.1). (B) Distribution of the ratio $\text{Res}(\mathcal{G})/\beta$ on the linear x -axis (0 to 4) over 400 graphs, with Axiom 3 holding (dark) and with one edge weight set to zero (light); the dashed line marks unity. The intact series lies entirely at or above 1; the falsified series places 109 of 200 graphs at 0. The bound is therefore a consequence of the axiom and not of the definition, since removing the axiom removes the bound. (C) Cut cost on the linear y -axis (0 to 20) versus the weight of the joining edge on the linear x -axis (0.2 to 34), for the two-triangle witness of Theorem 3.3 with all six internal weights fixed at 10. The minimum cut (solid) rises with the bridge while the best singleton cut (dashed) is constant at 20; the shaded interval is where the minimum is attained by a three-position region and by no singleton, and the dotted line marks the crossover at bridge weight 20 beyond which the two coincide. Individuation has extent (Theorem 3.4). (D) Three-dimensional surface of mean separation cost on the z -axis (4.08 to 35.3) over edge density on the x -axis (0.25 to 1.0) and position count $|V|$ on the y -axis (4 to 10), each cell the mean of 14 graphs. The surface is strictly positive at every combination and rises with both parameters; there is no direction in the parameter space along which separation cost approaches zero, which is the absence of the limiting process that Theorem 3.2 identifies as the theorem’s content. Viewing angle: elevation 24° , azimuth -58° .

14.6 Reproducibility

The prototype is implemented in Python with NumPy and SciPy (Harris et al., 2020; Virtanen et al., 2020) and graph routines from NetworkX (Hagberg et al., 2008); it is deterministically seeded and persists every experiment as a JSON record. Reproduction:

```
python prototype/runtime.py          # runtime self-test
python validation/run_validation.py  # full suite -> validation/results/
```

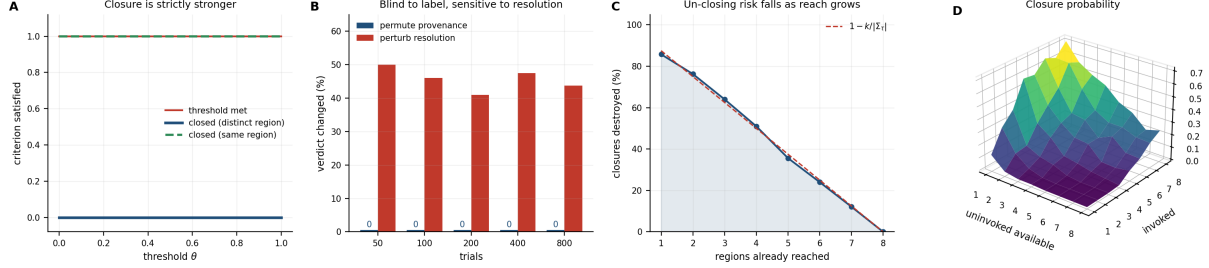


Figure 2: **Closure is not a threshold, does not read provenance, and is not monotone in what is available.** (A) Criterion satisfied, on the binary y -axis, versus the confidence threshold θ on the linear x -axis (0 to 1), on the witness of Theorem 4.5: two disjoint triangles joined by a single edge, with both considerations carrying confidence 1. The threshold criterion (upper trace) is met at every θ ; the closure criterion (lower trace) fails at every θ , because the uninvoked consideration resolves into the second triangle. Replacing that consideration by one resolving into the *first* triangle restores closure at every θ (dashed, coincident with the upper trace), locating the failure in the distinct region rather than in the form of Theorem 4.3. No adjustment of θ converts one criterion into the other. (B) Percentage of verdicts changed on the linear y -axis versus the number of randomised configurations on the categorical x -axis (50 to 800). Permuting provenance labels changes the verdict in exactly zero cases at every sample size (dark, annotated, drawn as a rule at the axis since the bars have no height); perturbing *resolutions* with the same machinery changes it in 41.0 to 50.0 per cent of cases (light). It is the asymmetry between the two counts, not the invariance alone, that distinguishes Theorem 4.12 from a definition too coarse to register anything: a definition insensitive to everything would have produced invariance under both perturbations. (C) Percentage of existing closures destroyed by adjoining one further available consideration, on the linear y -axis (0 to 85.8), versus the number of regions the invoked set already reaches, on the linear x -axis (1 to 8), drawn from a pool of eight distinct regions; the dashed line is the analytic prediction $1 - k/|\Sigma_\tau|$. The measured curve tracks it across the range and reaches zero exactly when the invoked reach exhausts the pool. A commitment can therefore lapse because something new became available, with no agent having changed its mind about anything already considered (Theorem 4.10). (D) Three-dimensional surface of the probability that a search is closed, on the z -axis (0 to 0.713), over the number of uninvoked available considerations on the x -axis (1 to 8) and the number invoked on the y -axis (1 to 8), each cell 160 draws from a six-region pool. The surface falls steeply along the availability axis and rises along the invoked axis: closure is a joint condition on both, and in particular is not a property of the evidence in hand (Theorem 4.9). Viewing angle: elevation 26° , azimuth -52° .

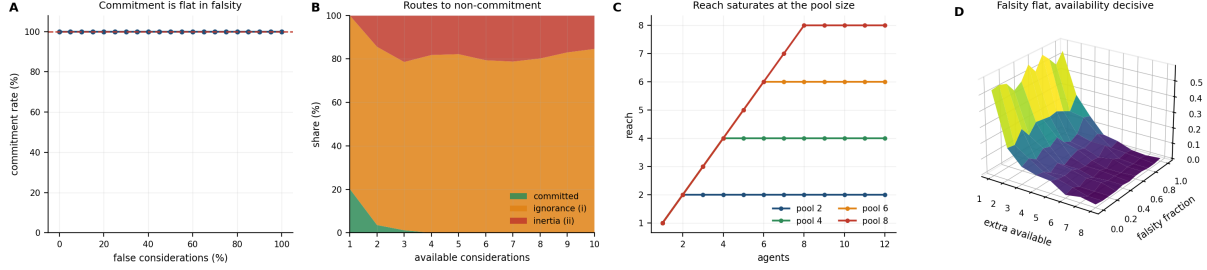


Figure 3: **Correctness is not a term in the commitment condition, and non-commitment decomposes into three structurally distinct routes.** (A) Commitment rate on the linear y -axis (0 to 108 per cent) versus the percentage of considerations that are false on the linear x -axis (0 to 100), holding resolutions fixed, 500 configurations per point; the dashed line marks 100 per cent. The measured rate is exactly 100 per cent at every level of falsity including the endpoint at which *every* consideration is false. Commitment is therefore invariant under substitution of false considerations for true ones with the same resolution (Theorem 4.16). The theorem does not assert that falsity is harmless: a population committed to the wrong region encounters the consequences of being there, but by the ordinary route of occupying an unsuitable region rather than through any failure of the mechanism (Theorem 4.17). (B) Share of outcomes on the linear y -axis (0 to 100 per cent) versus the number of available considerations on the linear x -axis (1 to 10), 500 configurations per point, stacked by the classification of Theorem 4.18. Commitment falls from 20.2 per cent at a single available consideration to below 1 per cent by four; ignorance — no invoked consideration reaches the target — dominates throughout at 77 to 82 per cent; inertia — the target is reached but closure fails — grows as the available set does. The three are distinguished by different tests, which is why they present differently from the inside (Theorem 4.21). (C) Reach on the linear y -axis (1 to 8) versus population on the linear x -axis (1 to 12), for pools of two, four, six and eight distinct resolutions. Each series grows one-for-one until the pool is exhausted and is exactly flat thereafter. Reach is bounded by the number of *distinct* resolutions and not by headcount, and the ceiling $|\Sigma_\tau(\mathcal{G})|$ counts regions the graph possesses, so no recruitment raises it (Theorems 4.23 and 4.24). (D) Three-dimensional surface of commitment probability on the z -axis (0 to 0.583) over the number of extra available considerations on the x -axis (1 to 8) and the falsity fraction on the y -axis (0 to 1), 120 draws per cell. The surface is flat along the falsity axis and falls sharply along the availability axis, exhibiting Theorem 4.16 and Theorem 4.10 on one pair of coordinates: what an agent believes wrongly does not move the verdict, and what it has not yet considered does. Viewing angle: elevation 24° , azimuth -56° .

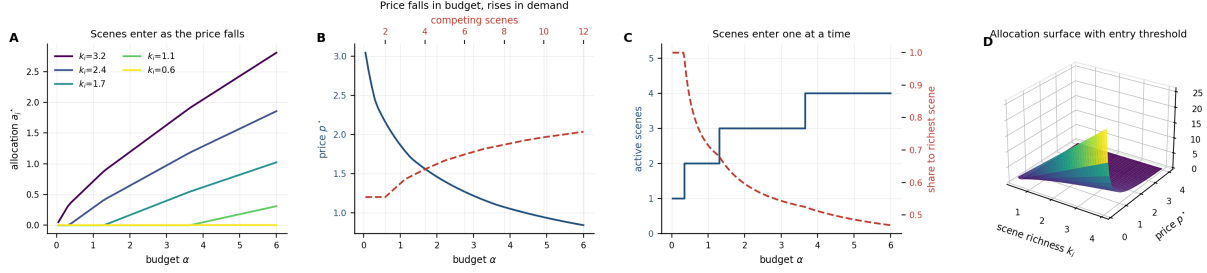


Figure 4: **A bounded capacity divides across concurrent demands by water-filling at a single price, with scenes entering at their own thresholds.** (A) Optimal allocation a_i^* on the linear y -axis (0 to 2.81) versus the budget α on the linear x -axis (0.05 to 6), for five scenes with gain profiles $\gamma_i(a) = k_i \log(1 + a)$ and richness $k_i \in \{3.2, 2.4, 1.7, 1.1, 0.6\}$. Each curve leaves zero at its own entry threshold — the budget at which the price falls to $\gamma_i'(0) = k_i$ — and the poorest scene remains unattended over the entire range plotted. Just above its threshold a scene receives a positive but arbitrarily small allocation, which is the formal minimal acknowledgement of Theorem 5.5. (B) Attention price p^* on the linear y -axis (0.84 to 3.05) versus the budget on the lower linear x -axis (0.05 to 6, solid) and versus the number of competing scenes on the upper linear x -axis (1 to 12, dashed, p^* from 1.20 to 2.04). The price is monotone decreasing in capacity and monotone increasing in demand, both as Theorem 5.3 requires of the multiplier of a binding budget constraint. (C) Number of attended scenes on the left linear y -axis (0 to 5, step) and the share of capacity received by the richest scene on the right linear y -axis (0.5 to 1.0, dashed), versus the budget on the linear x -axis (0.02 to 6). Scenes are taken up one at a time as the price crosses successive thresholds, while the richest scene’s share falls monotonically from unity: the agent becomes thinly present in several contexts rather than fully present in one. This is the derived content of Theorem 5.4, and it is the one result in the figure conditional on the environmental assumption Axiom 6; under increasing returns the optimum is concentration instead (Theorem 5.7). (D) Three-dimensional surface of the allocation $(\gamma_i')^{-1}(p^*) = k_i/p^* - 1$ on the z -axis (0 to 25) over scene richness k_i on the x -axis (0.4 to 4.0) and price p^* on the y -axis (0.15 to 4.0), clipped at zero where $k_i \leq p^*$. The ridge along which the surface leaves the zero plane is the entry threshold, and the divergence at small price is the regime in which an idle agent is forthcoming everywhere at once. Viewing angle: elevation 26° , azimuth -56° .

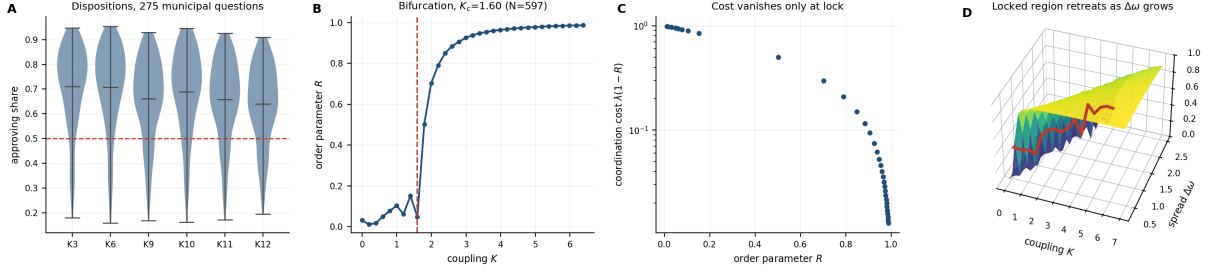


Figure 5: The synchronisation threshold, measured on the heterogeneity of the Zürich record. (A) Approving share on the linear y -axis (0.158 to 0.954) for each of the six consistently reported municipal units on the categorical x -axis, as violin densities over all 275 city questions of 2002–2026 (Statistik Stadt Zürich, 2026a); the dashed line marks the 0.5 majority. The dispersion between units, not their central tendency, is the velocity spread $\Delta\omega$ that fixes the threshold in the remaining charts, so the parameter is measured from the record rather than assumed. (B) Order parameter R on the linear y -axis (0.012 to 0.987) versus coupling K on the linear x -axis (0 to 6.38), with each unit treated as a tempo class populated in proportion to its resident count, giving $N = 597$ oscillators, integrated 3000 steps at $dt = 0.05$; the dashed line is the predicted critical coupling $K_c = 1.60$. Below it the order parameter remains at or under 0.107 and does not drift upward under continued integration; above $2K_c$ it exceeds 0.951. The populating step is necessary rather than cosmetic: Theorem 6.9(ii) is a mean-field statement and six oscillators alone leave the transition buried in finite-size fluctuation. (C) Coordination cost $\lambda_c(1 - R)$ on the logarithmic y -axis (0.0128 to 0.988) versus the order parameter on the linear x -axis (0 to 1), evaluated on the achieved R of chart B. The cost falls by nearly two decades only as R approaches unity and vanishes exactly at lock, which is the equality case of Theorem 6.6; away from lock it is strictly positive. (D) Three-dimensional surface of the order parameter on the z -axis (0 to 1) over coupling K on the x -axis (0 to 7) and velocity spread $\Delta\omega$ on the y -axis (0.3 to 2.5), computed on one fixed standardised velocity sample of 400 oscillators rescaled per row so that rows differ only in spread and not in sampling noise; the solid line is the predicted $K_c = 2\sigma_\omega\sqrt{2\pi}/\pi$ lifted onto the surface. The locked plateau retreats toward larger K as heterogeneity grows, and the predicted threshold tracks the boundary. The residual roughness below the threshold is genuine finite-size fluctuation of R in the incoherent regime, not numerical noise, and is left unsmoothed. Viewing angle: elevation 38° , azimuth -70° .

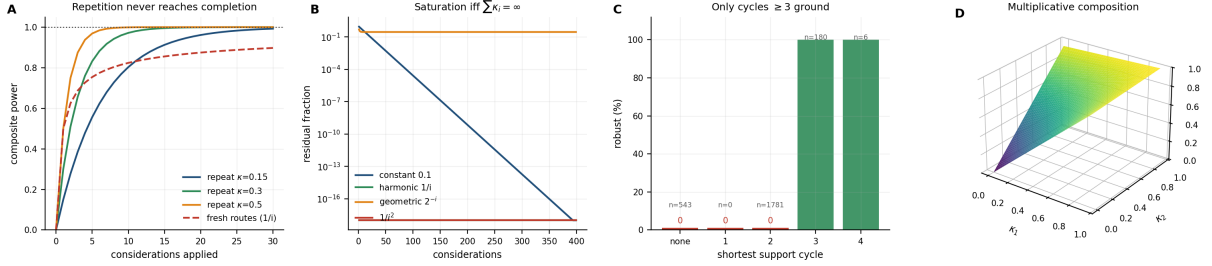


Figure 6: Considerations compose multiplicatively, repetition saturates strictly below completion, and grounding requires a support cycle of length at least three. (A) Composite power on the linear y -axis (0 to 1) versus the number of considerations applied on the linear x -axis (0 to 30), for a single consideration of power $\kappa \in \{0.15, 0.30, 0.50\}$ repeated (solid) and for a sequence of fresh routes with powers $0.5/i$ (dashed, reaching 0.897 at 30); the dotted line marks unity. Every repetition series increases, converges toward unity and attains it at no finite count, with strictly decaying increments ([Theorems 7.4](#) and [7.5](#)). (B) Residual fraction $\prod_i(1 - \kappa_i)$ on the logarithmic y -axis (10^{-18} to 1) versus the number of considerations on the linear x -axis (1 to 399), for four power sequences. The divergent-sum sequences (constant 0.1; harmonic $1/i$) drive the residual toward zero; the convergent-sum sequences (geometric 2^{-i} ; inverse-square $1/i^2$) plateau at a positive constant and remain there. This is the dichotomy of [Theorem 7.6](#) and the structural condition on a long campaign: a proposal whose reachable considerations have summable powers is never closed by persistence ([Theorem 7.7](#)). (C) Percentage of support structures that are robust on the linear y -axis (0 to 100) by shortest directed cycle length on the categorical x -axis, exhaustively over four-position digraphs with at most six edges, with the number of structures in each class annotated. Acyclic structures and cycles of length one and two are robust in no case and are drawn as rules at the axis with their zero marked; every structure whose shortest cycle has length three or four is robust without exception. Two elements can only assert each other; three can check each other ([Theorems 7.9](#) and [7.10](#)), and a repeated single item is a one-cycle and therefore grounds nothing ([Theorem 7.13](#)). (D) Three-dimensional surface of the composite power $1 - (1 - \kappa_1)(1 - \kappa_2)$ on the z -axis (0 to 1) over the two constituent powers on the x - and y -axes (0 to 0.95). The sheet rises toward but never attains unity, and its curvature is the diminishing return each additional consideration closes of the remaining above-floor distance ([Theorem 7.3](#)). Viewing angle: elevation 26° , azimuth -54° .

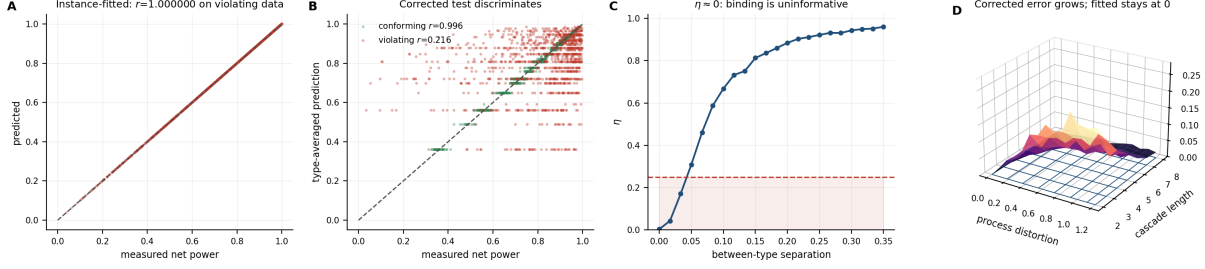


Figure 7: The natural test of the composition law has no null; the type-averaged test does; and the diagnostic says when even it has no power. (A) Instance-specific prediction $1 - \prod_i(1 - \kappa_i)$ on the linear y -axis (0 to 1) versus the measured net power $(S_1 - S_{n+1})/(S_1 - \beta)$ on the linear x -axis (0 to 1), for 1500 cascades of length two to six generated by a process that violates [Theorem 7.3](#) by construction, each step’s reduction being scaled by an arbitrary distortion factor. The points lie on the diagonal (dashed) with Pearson $r = 1.000000$: under instance-specific estimation the two quantities are the *same algebraic expression*, so the statistic attains its degenerate value on every data set whatsoever, including data generated by a process violating the law it purports to confirm. The test cannot fail and therefore has no power against any alternative ([Theorems 12.2](#) and [12.3](#)). (B) The same comparison under type-averaged estimation, in which the constituent powers are the means of their event types computed excluding the predicted cascade. Conforming data attain $r = 0.996$ and data violating the composition law fall to $r = 0.216$, with the violating cloud departing visibly from the diagonal. Breaking the dependence of the prediction on the predicted cascade’s own observations restores a null ([Theorem 12.6](#)). (C) Separation diagnostic $\eta = \text{Var}_b / (\text{Var}_b + \text{Var}_w)$ on the linear y -axis (0.004 to 0.961) versus the imposed between-type separation on the linear x -axis (0 to 0.35), at fixed within-type dispersion; the shaded band below the dashed line at 0.25 is the regime in which per-type means are indistinguishable. There, predictions are constant at fixed cascade length and no comparison among composition rules can be separated, so a negative result is evidence about the observable rather than about the process ([Theorem 12.8](#)). (D) Three-dimensional mean absolute prediction error on the z -axis over process distortion on the x -axis (0 to 1.2) and cascade length on the y -axis (2 to 8), 60 cascades per cell, for the type-averaged test (surface, rising to 0.247) and the instance-specific test (wireframe, bounded by 5.7×10^{-17} across the whole grid). The corrected error grows with distortion while the fitted error is pinned at floating-point zero for every process: the obstruction and its remedy in one view. Viewing angle: elevation 24° , azimuth -58° .

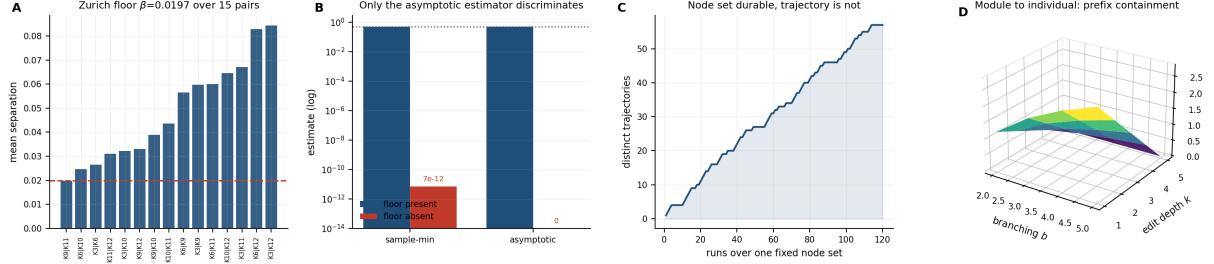


Figure 8: **The measured floor of the Zürich binding, the two discharges of the floor obligation, and the runtime's two durabilities.** (A) Mean separation on the linear y -axis (0.0197 to 0.0844) for each of the 15 receiver pairs on the categorical x -axis, sorted ascending, computed as the mean absolute difference of approving shares over all 275 questions; the dashed line marks the least pair separation, which is the measured floor $\beta = 0.0197$. It is strictly positive: even the two most similar units differ by about two percentage points on average across the record, so the floor is measured rather than assumed. Published shares are rounded to 0.1 percentage points and two units therefore coincide exactly on 30 of 4125 single questions (0.73 per cent); a floor defined per question would be zero for a reason having nothing to do with the polity, which is why obligation (S4) is discharged on receiver *pairs* over the whole record, as [Theorem 2.2](#) requires. (B) Floor estimate on the logarithmic y -axis (10^{-14} to 3) for the two discharges on the categorical x -axis, evaluated on a synthetic process with a genuine floor at 0.5 (dark) and on one constructed to have no floor (light), 4000 observations each; the dotted line marks the true value 0.5. The sample-minimum estimator returns 0.500 and 7×10^{-12} — positive on *both* processes, so a positivity claim from it carries no evidential weight; the asymptotic estimator returns 0.499 and exactly 0, discriminating between them and therefore capable of being wrong ([Theorem 9.4](#)). (C) Cumulative count of distinct trajectories on the linear y -axis (1 to 57) versus runs executed on the linear x -axis (1 to 120), over one fixed six-node set. The node set is durable and can be catalogued, frozen and imported; the induced causal edge relation is constituted by the run and is not schedulable in advance, so reproducibility attaches to the protocol and its provenance rather than to the readings ([Theorems 11.13](#) to [11.15](#)). (D) Three-dimensional surface of the blast radius b^{D-k} on the logarithmic z -axis (10^0 to $10^{2.6}$) over branching factor b on the x -axis (2 to 5) and edit depth k on the y -axis (1 to 5), at fixed tree depth $D = 5$. An edit at depth k touches exactly the leaves beneath its length- k prefix and nothing else, falling from an entire imported subtree to a single leaf. A module is a short prefix and an individual a full-depth address in one space, so producing an individual by pruning is descent in that space rather than a change of kind ([Theorems 9.7](#) and [9.8](#)). Viewing angle: elevation 24° , azimuth -56° .

Per [Theorem 11.14](#) what is reproduced is the protocol and its provenance; per [Theorem 8.11](#) repeated *sessions* need not agree, and their differences are data.

Part V — Scope

15 Ethical commitments

The instrument generates agents from public records and places human participants in extended interaction with them. Four commitments follow, and each is enforced in the implementation.

Modules are patterns, not persons. By [Theorem 9.1](#) a module is a τ -sufficient region — a class of disposition. No module is derived from, or intended to represent, an identified individual, and pruning ([Theorem 9.5](#)) yields fictional agents consistent with aggregate statistics. This is not merely a policy: by [Theorem 3.4](#) the object individuated at any resolution is a region, so a claim to have reconstructed a person would be a claim the formalism does not support.

Participants are told what the instrument cannot do. By [Theorem 12.13](#) the system cannot report which of a participant’s actions mattered. Presenting any readout that appears to do so would misrepresent the instrument; [Invariant 6](#) forbids it structurally.

Deposits are irreversible and disclosed. By [Theorem 5.12](#) nothing a participant does can be withdrawn from an agent’s record. Participants should be informed of this before a session, since it is unusual.

No claim is made about the real city. The binding of [Section 14.2](#) uses Zürich’s public record as a source of vocabulary and heterogeneity. Statements produced by agents are statements about the simulated population and are not assertions about Zürich, its institutions, or its residents.

16 Limitations

We record what the development does not establish.

The environmental axiom. [Theorem 5.3](#) and everything depending on it hold only under [Axiom 6](#), a claim about the world rather than the agent. Where scenes offer increasing returns the optimum is concentration, not division ([Theorem 5.7](#)). All results of [Sections 3, 4 and 7](#) and [Theorems 5.8, 5.12, 8.3 and 8.8](#) are independent of it.

Availability. [Theorem 4.3](#) quantifies over an available set the theory does not determine ([Theorem 4.7](#)). Closure is decidable given availability and not from the graph alone. The instrument therefore cannot say whether a population *ought* to be committed to something.

Empirical adequacy. We do not claim the agents model human deliberation. The Zürich binding supplies realistic heterogeneity, not validated behaviour, and no experiment reported here compares agent conduct to human conduct. Establishing or refuting such a correspondence is the natural next study and would leave every theorem intact, since none asserts it.

The independence premise. [Theorem 10.5](#) enforces a *declared* independence relation. The system cannot verify that considerations are genuinely independent ([Theorem 10.8](#)), and three considerations sharing an upstream source satisfy the rule while violating its intent.

Scale. [Invariant 4](#) requires the graph to grow monotonically, and [Theorem 5.12](#) forbids removing committed record. Compaction consistent with these invariants — summarising an agent’s

structure while preserving (χ, m) — is not developed here and is the principal engineering problem at population scale.

Bands and thresholds. The only genuine transition in the model is the bifurcation of [Theorem 6.9](#). Any descriptive banding of the order parameter is a classification cut-off on a continuous variable and carries no claim of criticality.

17 Conclusion

We set out to specify an instrument in which a well-attested but hard-to-study social fact — a proposal believed feasible, opposed by no one, and never adopted, with no participant experiencing a decision — can be instantiated and perturbed.

The explanation the formalism supplies is a single structural observation in two halves. Commitment is *closure*: an agent commits when every available consideration resolves into a region already reached, a condition strictly stronger than any confidence threshold ([Theorem 4.5](#)). And closure is *blind*: the condition quantifies over resolutions and mentions nothing else, so it is invariant under relabelling of provenance ([Theorem 4.12](#)) and of truth ([Theorem 4.16](#)). The proposal is not rejected; its region never closes, and a failure of closure is not an event ([Theorem 4.20](#)). At population scale the same absence recurs as a stable sub-threshold state ([Theorem 6.9](#)), and in the algebra of considerations as a residual that plateaus above the floor ([Theorem 7.6](#)).

The instrument that follows has an unusual measurement surface, and its refusals are its principal design commitments rather than concessions. It cannot report success, because the comparison an exit code encodes is not computable in it ([Theorem 11.7](#)). It cannot report which consideration mattered, because that verdict is a global minimum expressed in local terms and no store of local content expresses it ([Theorem 12.10](#)), and because the terminus does not determine the path ([Theorem 11.16](#)). And the natural test of its own composition law is an identity with no null ([Theorem 12.2](#)), so the instrument exposes only the corrected, type-averaged test together with the diagnostic that says when even that has no power ([Theorem 12.8](#)).

What remains is considerable: a complete and interrogable record of what propagated, ordinal verdicts on the structure of an argument ([Theorem 7.11](#)), distributional signatures across a heterogeneous population, and a protocol whose provenance is reproducible even though its readings are not ([Theorem 11.15](#)). We regard the combination — complete visibility of what happened, principled inability to determine what mattered — as the honest measurement surface for this class of phenomenon, and note that a version of the instrument reporting an effect attribution would be evidence that the theory it implements is wrong.

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